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"Vegetation and Energy"

"CGLOPS-1"

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List of Acronyms

AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
BA	Burned Area
C3S	Copernicus Climate Change Service
CAL/VAL	Calibration and Validation
CCI	Climate Change Initiative
Ce	Commission error rate
CEOS	Committee on Earth Observation Satellite
CGLOPS	Copernicus Global Land Operations
CLMSBA40	Burnt area v4.0 product of CLMS
CLMS	Copernicus Land Monitoring Service
COG	Cloud Optimized Geotiff
DC	Dice coefficient
ECV	Essential Climate Variable
ESA	European Space Agency
Fire_cci	ESA Fire Disturbance project
GBOV	Global Land Ground Observation Framework Contract
GCOS	Global Climate Observing System
INSPIRE	Infrastructure for Spatial Information in the European Community
JRC	Joint Research Centre
LPV	Land Product Validation
MODIS	Moderate Resolution Imaging Spectrometer
MODIS-MCD64	MODIS Collection 6 Global monthly burned area product Level 3 gridded 500 m
NASA	National Aeronautics and Space Administration
NETCDF	Network Common Data Form
NIR	Near infrared
NRT	Near-real time
NTC	Non-Time Critical
Oe	Omission error rate
OLCI	Ocean and Land Color Instrument
PQAD	Product Quality Assurance Document
PUM	Product User Manual
QAR	Quality Assessment Report
relB	Relative Bias
RMSE	Root mean square error
S2GM	Sentinel-2 Global Mosaic
S3	Sentinel-3
SLSTR	Sea and Land Surface Temperature Radiometer

TLS	Total least squares
TSA	Thiessen Scene Area
VIIRS	Visible Infrared Imaging Radiometer Suite
WGS	World Geodetic System

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. The production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

The CLMS portfolio includes the Version 4.0 of the near-real time (NRT) and non-time critical (NTC) Burned Area (BA) products. The NRT product provides BA estimates within one day after each new image acquisition, as opposed to the NTC product that is available several weeks after image sensing but with an expected higher accuracy.

The current Validation Report presents the intercomparison and validation results as planned in the Product Quality Assurance Document [CGLOPS1_PQAD]. The analysis includes the Version 4.0 of the CLMS NRT and NTC Burned Area products (CLMBA40nrt and CLMBA40ntc, respectively) as well as the MODIS-MCD64 Collection 6 (MCD64) and the ESA Fire Disturbance project FIRECCIS311 products. The validation analysis is based on reference datasets generated in the framework of Copernicus Climate Change Service (C3S). This reference datasets consist of BA maps derived from supervised classifications of consecutive pairs of Landsat images at 106 scenes selected with a stratified random sampling in 2020.

The intercomparison and validation analyses show evidence of the product's quality and its consistency on a multi-year basis. The accuracy of CLMBA40ntc (DC 65.0%) is clearly higher than any other product, and the accuracy of CLMBA40nrt (DC 56.5%) is similar to the non-CLMS product with the highest accuracy, FIRECCIS311 (DC 56.0%). CLMBA40nrt has the clear advantage of providing BA estimates every day at near-real time instead of several months after image sensing.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The document presents the results of the scientific validation of the near-real time (NRT) and non-time critical (NTC) products of Version 4.0 of the Burned Areas (BA) algorithm. They are referred to as CLMBA40nrt and CLMBA40ntc, respectively.

The two CLMS Version 4.0 products are derived from Ocean and Land Colour Instrument (OLCI) and Sea and Land Surface Temperature Radiometer (SLSTR) data, onboard the Sentinel-3 platform, and an active fire product, the NASA's VIIRS Active Fire and Thermal Anomalies product.

The quality assessment includes (1) an intercomparison analysis over five years, 2019 - 2024, involving two external global NTC products apart from CLMS BA products, and (2) a validation analysis for year 2020, based on independent reference data derived from time series of Landsat imagery.

The objective of this report is to describe the quality of CLMBA40nrt and CLMBA40ntc products.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the requirements, and the expected performance
- Chapter 3 describes the methodology of quality assessment analyses
- Chapter 4 presents the results of the analyses
- Chapter 5 discusses the results
- Chapter 6 presents the conclusion of the quality assessment
- Chapter 7 makes recommendations based upon the results and conclusions
- Chapter 8 lists the references

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document for Vegetation and Energy products of the Copernicus Global Land Service
CGLOPS1_ATBD_BA300m-V4.0	Algorithm Theoretical Basis Document of the Burned Areas Version 4.0 product.

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output

Document ID	Descriptor
CGLOPS1_PUM_BA300m-V4.0	Product User Manual of the Burned Areas Version 4.0 product.

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External document

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

Fire_cci_D4.1.1_PVR_v2.1	ESA CCI ECV Fire Disturbance: D4.1.1. Product Validation Report, Version 2.1, 22nd December 2018.

Available at https://climate.esa.int/media/documents/Fire_cci_D4.1.1_PVR_v2.1.pdf

2 REQUIREMENTS

According to the applicable document [AD1], the requirements relevant for the burned area product are inherited without any modification from the GCOS#245 document. Some of the terms can be misleading, particularly "uncertainty" in item "Uncertainty (2-sigma)" refers to "accuracy" given the objective stated in the item description.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300m
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global

BA -BURNT AREAS	
Burned area is described by a grid where each cell is labelled as burnt if the majority of that cell is classified as containing burned vegetation [m]. (GCOS#245)	
Time span	Monthly
Timeliness	ASAP within 3 months after observation
Uncertainty (2-sigma)	Threshold: 25% Goal: 5%
Stability (per decade, i.e. 10 years)	Threshold: 2% Goal: 1%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NetCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

2.2.5 Input data

Copernicus Sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, PROBA-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 QUALITY ASSESSMENT METHODS

The quality assessment is based on intercomparison and validation analyses, as detailed in the Product Quality Assurance Document [CGLOPS1_PQAD].

The current quality assessment includes the two Version 4.0 CLMS products, one generated at near-real time (NRT) and the other generated at non-time critical (NTC) basis (they are referred to as CLMBA40nrt and CLMBA40ntc, respectively).

Additionally, for benchmarking purpose, two external global BA products are included in the assessment: the MODIS-MCD64 Collection 6 product (Giglio et al., 2018) (hereinafter referred to as MCD64); and the ESA's Climate Change Initiative Fire Disturbance project FIRECCIS311 product, derived from Sentinel-3 OLCI&SLSTR data (Lizundia-Loiola et al., 2022). For the sake of brevity, the Burnt Area product of the Climate Change Service (C3S) is not included in this report as it was found to be less accurate than FIRECCIS311 (Lizundia-Loiola et al., 2022).

3.1 INTERCOMPARISON METHOD

The intercomparison analysis consists of assessing the agreements between each evaluated product against FIRECCIS311, as its previous version, FIRECCIS310, was found to be the most accurate external product on a validation exercise based on Landsat-derived reference data (Franquesa et al., 2022). The analysis is undertaken using spatio-temporal units defined temporally by calendar months and spatially by 0.25º grid cells, the temporal and spatial resolutions common in BA grid products, as in the Global Fire Emissions Database (Giglio et al., 2013) and in the grid global product versions of FIRECCIS311 (Lizundia-Loiola et al., 2022).

For each product, the spatial extent of burned, unburned and no-data areas is aggregated at each monthly grid cell. The product raster pixels are vectorized and reprojected to the Universal Transverse Mercator coordinate system, and the spatial extents are then calculated as planar areas (in m^2). The total monthly BA at a grid cell, a_b, is estimated from the area with observed burned detections (as reported by the product), $a_{o,b}$, accounting for the area with available observations, a_o, and the total area of the grid cell, a_t (burned, unburned or no-data). This avoids attributing a low amount of BA to a grid cell simply because most of its area is unobserved.

$$a_b = \frac{a_{o,b} a_t}{a_o} \quad (1)$$

The agreement between products is assessed with linear regression metrics, the root mean square error (RMSE), similarly as in Humber et al. (2019), and with the coefficient of determination (r^2) as

in (Chuvieco et al., 2022). The slope and the line's offset, fitted with total least squares (TLS), indicate relative bias between the two products. The RMSE and the r^2 indicate the dispersion between the two products or the tendency to disagree in the amount of BA across the monthly grid cells.

3.2 VALIDATION METHOD

3.2.1 Reference data

The validation is based on a reference dataset for year 2020 generated by the Copernicus Climate Change Service (C3S) kindly provided by the C3S team from the University of Alcalá. Similar datasets are available for years 2017-2019, publicly available at the e-cienciaDatos repository (https://doi.org/10.21950/BBQQU7; Franquesa et al., 2020). Note that CLMS BA products are not available for 2017 and 2018 since they are derived from Sentinel-3 OLCI and SLSTR observations. Furthermore, the 2019 reference dataset was used for the algorithm calibration. The methods to generate the reference data, the sampling design, and the accuracy inference formulae are detailed in Franquesa et al. (2022). They are fundamentally similar to previous CLMS BA and ESA Fire Disturbance project validation exercises [Fire_cci_D4.1.1_PVR_v2.1]. The reference burned area maps were generated from pairs of Landsat images, separated by up to 32 days, by a semi-automatic classification algorithm, a random forest classifier, supervised by an expert interpreter. The classification process was repeated by refining the training areas until the interpreter detected no errors (Franquesa et al., 2022). Long units cover time periods between 48 days and up to a year, more than 110 days in 50% of the cases.

Accuracy inferences are based on stratified combined ratio estimator (Cochran, 1977), taking into account the size of the sampling units. The main difference with the Franquesa et al. (2022) methods is in the meaning attributed to products' no-data. Pixels with no-data in the product or in any of the reference images are excluded from the analyses and are flagged as no-data in the long unit reference map.

The accuracy estimated with long units is expected to be higher than with short units (made from single image pairs) (Franquesa et al., 2022; [Fire_cci_D4.1.1_PVR_v2.1]). The long units allow for larger temporal overlaps between product and reference datasets, what decreases the impact of temporal burn detection errors on the confusion matrix.

3.2.2 Sampling design

The stratification of the sampling design was based on (1) calendar years, (2) biomes derived from the Ecoregions 2017 map (Dinerstein et al., 2017) (Figure 1), and (3) low and high fire activity regions (Table 1). For the last stratification level, each year and biome were split in two according to its 80th percentile of BA. The annual sample size (106 TSAs for 2020) was allocated across strata

proportionally to stratum size (in number of sampling units) times the square root of BA, as recommended by Padilla et al. (2017a). The small sample sizes of validation data in most biomes, together with common relatively large variability of accuracies at biome level, pose a question on the validity of per-biome accuracy inferences (Franquesa et al., 2022; Padilla et al., 2017b). Therefore, per-biome accuracy results are not included in this report.

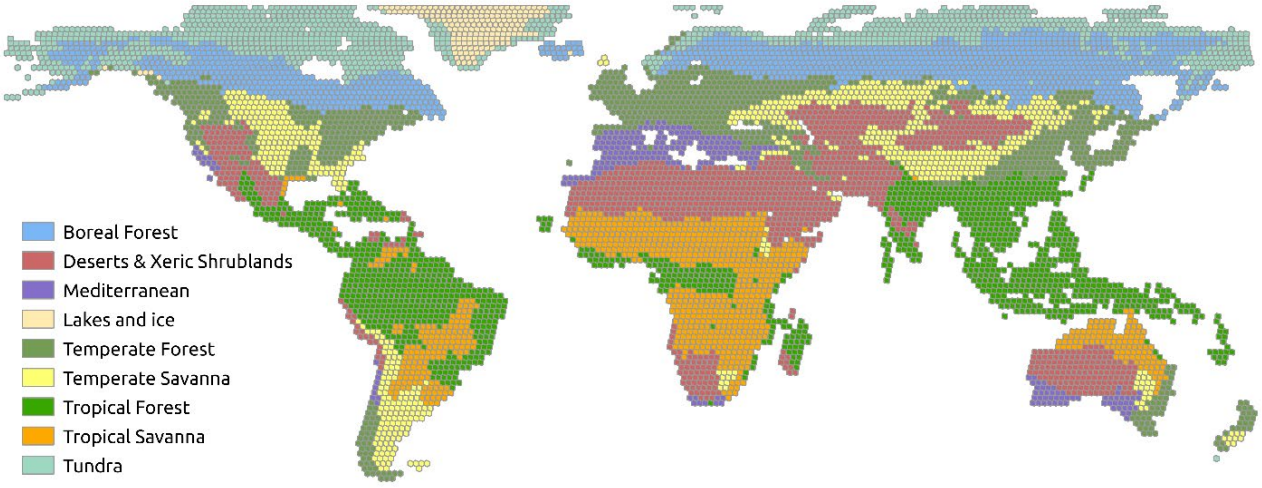


Figure 1: Biomes stratification derived from the Dinerstein et al. (2017) Ecoregions 2017 map (Franquesa et al., 2022).

Table 1: Sample size across strata for 2020.

Biome	Fire activity	
	high	low
Boreal Forest	4	2
Deserts & Xeric Shrublands	6	2
Mediterranean	2	2
Temperate Forest	4	2
Temperate Savanna	4	4
Tropical Forest	12	6
Tropical Savanna	21	31
Tundra	2	2

The size of each sampling unit and its associated subsample in Franquesa et al. (2022) is defined only with the spatial dimension, which impedes taking into account the variations in the temporal length

of available data. Long units were designed as one-year-long units. However, large cloud coverages often occur in at least one of the reference images, which lead a split of the unit on the temporal dimension and shorter periods with available data. This leads to an underestimation of estimates of totals but it should have a limited impact on the accuracy measures (estimates of ratios). A possible solution could be to define the sampling unit and subsample size with the spatial and temporal dimensions, by multiplying area and time, as defined in one of ESA Fire Disturbance project validation reports [Fire_cci_D4.1.1_PVR_v2.1]. However, a method published in a peer-reviewed scientific journal should be prioritized for the present report. Hence, the option proposed by Franquesa et al. (2022) is used here.

3.2.3 Accuracy estimates

Accuracy is commonly characterized in BA validation exercises by indices derived from an error matrix, which quantifies agreement and disagreement in terms of area (m^2) between product and reference classifications (Padilla et al., 2017a) (Table 2). Products' no-data was excluded from the error matrices, similarly as in Padilla et al. (2015, 2014) and Cochran (Cochran, 1977; Section 5A14). Furthermore, one of the validation units in the 2020 sample falls outside the spatial coverage of MCD64, located slightly above the product's northern coverage limit, at a latitude of 70º north. Therefore, accuracy inferences are based on the 105 units with available data for all products, to ensure fair comparisons between the products' accuracies. Accuracies can be estimated by considering the subpopulation or "domain" of sampling units that fall within the common products' spatial coverage. This can be achieved by modifying the definition of the error matrix cells, setting them to zero for units outside the subpopulation of interest, in this case, for this unit above latitude 70º north.

Table 2: Population error matrix. Matrix cells express agreement (diagonal cells) or disagreement (off-diagonal cells) in terms of the area between the BA product (map) class and the reference classification.

	Reference classification		
Product classification	Burned	Unburned	Row total
Burned	E_{11}	E_{12}	E_{1+}
Unburned	E_{21}	E_{22}	E_{2+}
Column Total	E_{+1}	E_{+2}	

Accuracy measures are commonly ratios of combinations of error matrix entries. For example, for the "burned" class, the commission error ratio (Ce) is

$$Ce = \frac{E_{12}}{E_{1+}} \tag{2}$$

and the omission error ratio (Oe) is

$$Oe = \frac{E_{21}}{E_{+1}} \tag{3}$$

where E_{ij} refers to the population values of the error matrix entries (Table 2). Additionally, accuracy can be also measured with the Dice coefficient (DC) (Dice, 1945), which is particularly useful for comparing product accuracies (Franquesa et al., 2022; Padilla et al., 2015) as it combines both error ratios (*Ce* and *Oe*) into a single measure. DC has a sensible probabilistic interpretation, it is defined as the conditional probability that one classifier identifies a pixel as burned, given that the other classifier also identified it as burned (Fleiss et al., 2013).

$$DC = \frac{2E_{11}}{2E_{11} + E_{12} + E_{21}} \tag{4}$$

Product bias is measured as the difference between the total area burned detected by the product versus the reference map, standardized by the total area burned detected by the reference map.

$$relB = \frac{E_{1+} - E_{+1}}{E_{+1}} \tag{5}$$

3.3 ANALYSED SATELLITE PRODUCTS

- CLMS BA NTC product Version 4.0 (CLMBA40ntc):
  CLMBA40ntc is an NTC product derived from CLMS 300m Sentinel-3 OLCI&SLSTR surface reflectances Version 2.3 and NASA's VIIRS Active Fire and Thermal Anomalies product. The product is based on semi-empirical models calibrating a deep learning algorithm, consisting of long short-term memory and convolutional neural networks, specifically designed to process time series of reflectance data. The derived time series of burned area detection maps are revised by active fire detections spatiotemporal densities to filter out areas likely to be related to land surface changes other than fires, such as agricultural harvests or ploughing, or fast vegetation senescence. The algorithm was calibrated with active fire detections and the C3S reference dataset (Franquesa et al., 2022) for 2019. The product is generated several weeks after image sensing and, therefore, with time series of reflectance data after each mapping day. Like the other NTC products, this product is delivered as

monthly temporal mosaics [CGLOPS1_ATBD_BA300m-V4.0]. It is currently available from July 2018 to December 2024.

- CLMS BA near-real time (NRT) product Version 4.0 (CLMBA40nrt):
  CLMBA40nrt is an NRT product derived from the same algorithm and input reflectances as in CLMBA40ntc, but it is generated with each new Sentinel-3 image; hence, it does not use images after each mapping day, and it is delivered as daily files [CGLOPS1_ATBD_BA300m-V4.0]. It is calibrated with 2019 data, similarly as in the NTC algorithm. It is available for 2020 (internal product) and from 1st December 2024 to the present. The quality assessment uses data up to August 2025.
- MODIS-MCD64 Collection 6 product (MCD64):
  MCD64 is an NTC product derived from 500m MODIS data and active fire detections from the Collection 6 Terra MOD14A1 and Aqua MYD14A1 products. The product algorithm first generates images of persistent changes of a burn-sensitive vegetation index, which are then classified as burned / unburned with probabilistic thresholds drawn from samples at active fire detections (Giglio et al., 2018). It is currently available from November 2000 to December 2024.
- ESA Climate Change Initiative Fire Disturbance project (FIRECCIS311):
  FIRECCIS311 is an NTC product derived from 300m Sentinel-3 SLSTR data and active fire detections from the NASA's VIIRS Active Fire and Thermal Anomalies product. The retrieval methodology uses a contextual growing algorithm based on shortwave infrared reflectance thresholds dynamically selected on each processing 10ºx10º tile (Lizundia-Loiola et al., 2022). It is currently available from January 2019 to December 2022.

4 RESULTS

4.1 INTERCOMPARISON ANALYSIS

Figures 2-5 show the spatial distribution of CLMBA40ntc, CLMBA40nrt, FIRECCIS311 and MCD64 BA estimated for 2020, the year with available data for all products. Annex 1 includes the same figures but for the annual average burned area over the period 2019-2024 for CLMBAntc, and MCD64 and over the period 2019-2022 for FIRECCI311. Annex 2 shows the same figures for 2020 but disaggregated by month. Figures 6-7 show the difference maps for 2020 between CLMBA40 products and FIRECCIS311, Figures 8-9 between CLMBA40 products and MCD64, and Figure 10 between CLMBA40ntc and CLMBA40nrt.

The spatial distributions of BA estimates of the four products are remarkably coincident. High amounts of BA are detected by all products in African tropical savannahs and in the Northern Australia, South America, Central Asia, South East Asia and Boreal regions. CLMBA40 products tend to estimate more BA than non-CLMS products (red areas on Figure 6, Figure 7, Figure 8 and Figure 9), but mainly on regions where the later products already estimate large amounts of BA. CLMBA40ntc tend to estimate more BA than CLMBA40nrt in some agricultural lands, mainly in north-east India, but also in East Asia and East Europe, and also in some regions of African tropical savannas (blue areas on Figure 10). Exploratory analyses of these discrepancies, illustrated in Annex 3, indicate that CLMBA40ntc is slightly more sensitive to land surface changes than the other products, leading to correct detections of burned area patches partly detected by the other products. The higher BA estimates in CLMBA40nrt than in CLMBA40ntc (red areas on Figure 10) tend to correspond to commission errors due to residual cloud shadows (not detected by the Sentinel-3 quality layers). Specifically, over agricultural lands in north-east India, CLMBA40nrt and MCD64 tend to estimate much less BA than the other two products. The exploratory analyses indicate that those discrepancies occur at a complex region with a mosaic of agricultural harvests&ploughing and agricultural fires, where their discrimination is even difficult with a visual inspection of Sentinel-2 imagery and active fire detections. Other discrepancies in tropical regions in central Africa correspond to commission errors, mainly in FIRECCIS311, and omission errors in CLMS products (blue areas on Figures 6-7). Relatively large clusters of active fires are detected in regions with known small-scale deforestation processes taking place (Butsic et al., 2015). However, while reflectance changes are not present at the Sentinel-3 time series, they are present at Sentinel-2 20 m observations but not at the locations of the FIRECCIS311 detections.

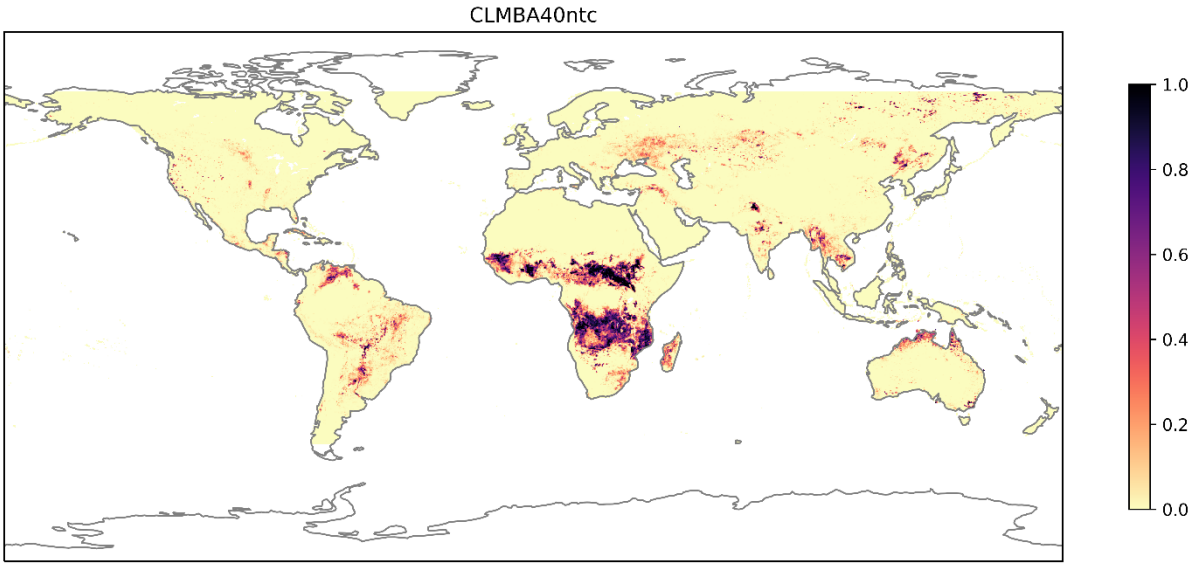


Figure 2: Proportion of burned area in each 0.25º grid cell for 2020 according to CLMBA40ntc. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell, in any of the product's monthly composites (Annex 2).

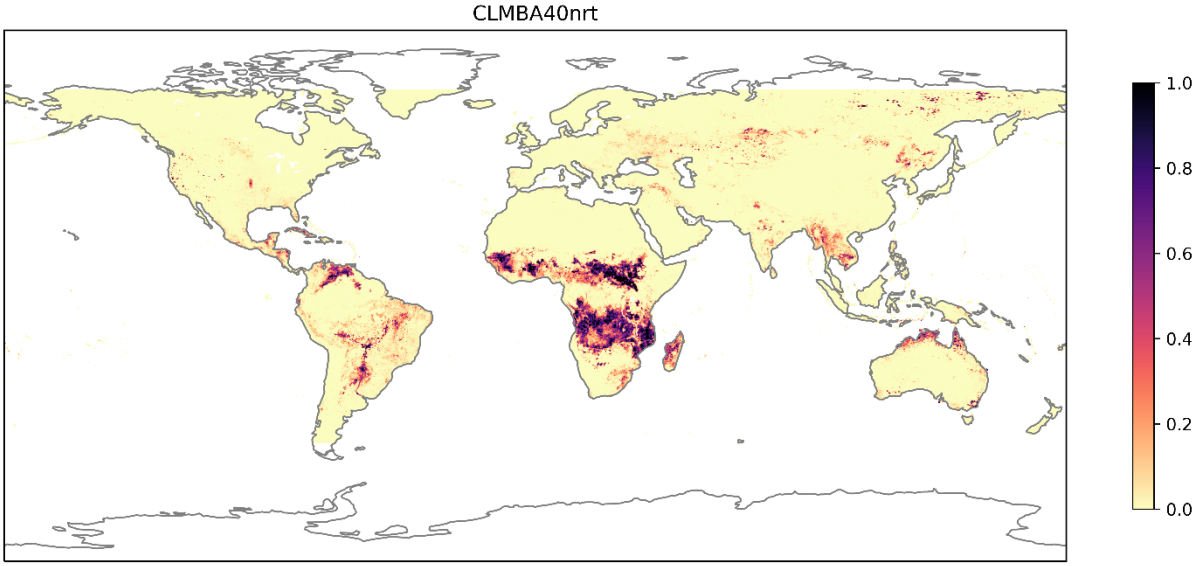


Figure 3: Proportion of burned area in each 0.25º grid cell for 2020 according to CLMBA40nrt. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell, in any of the product's monthly composites (Annex 2).

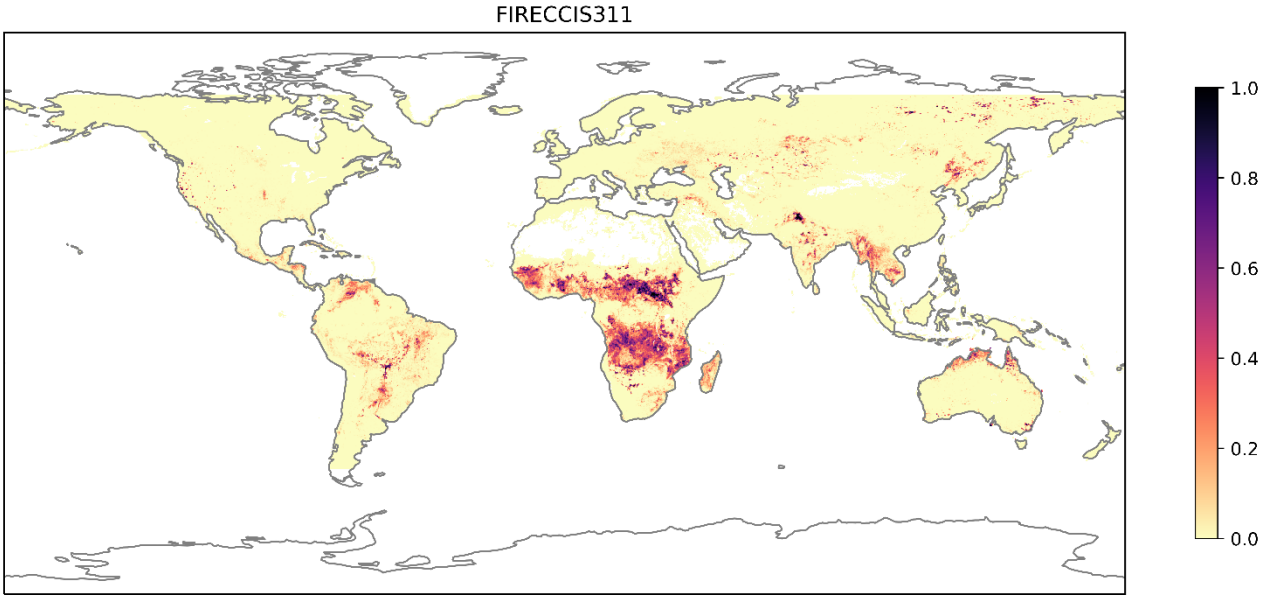


Figure 4: Proportion of burned area in each 0.25º grid cell for 2020 according to FIRECCIS311. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell, in any of the product's monthly composites (Annex 2). White additionally represents non-burnable areas (e.g., bare areas and permanent snow and ice) flagged similarly as permanent water in this product.

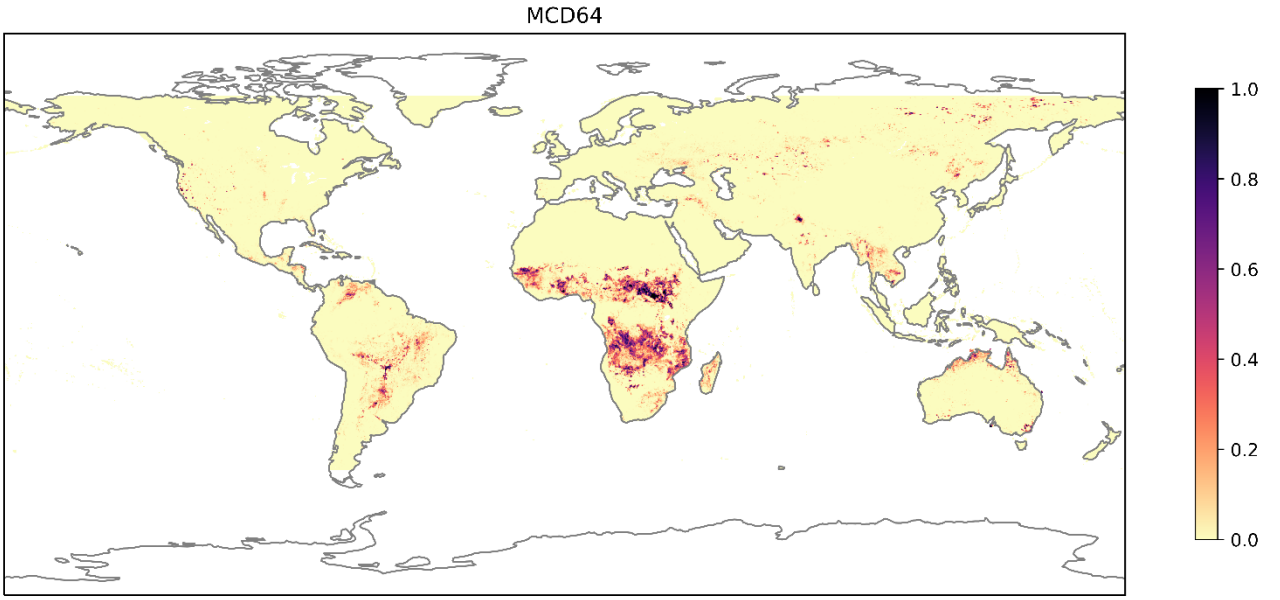


Figure 5: Proportion of burned area in each 0.25º grid cell for 2020 according to MCD64. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell, in any of the product's monthly composites (Annex 2).

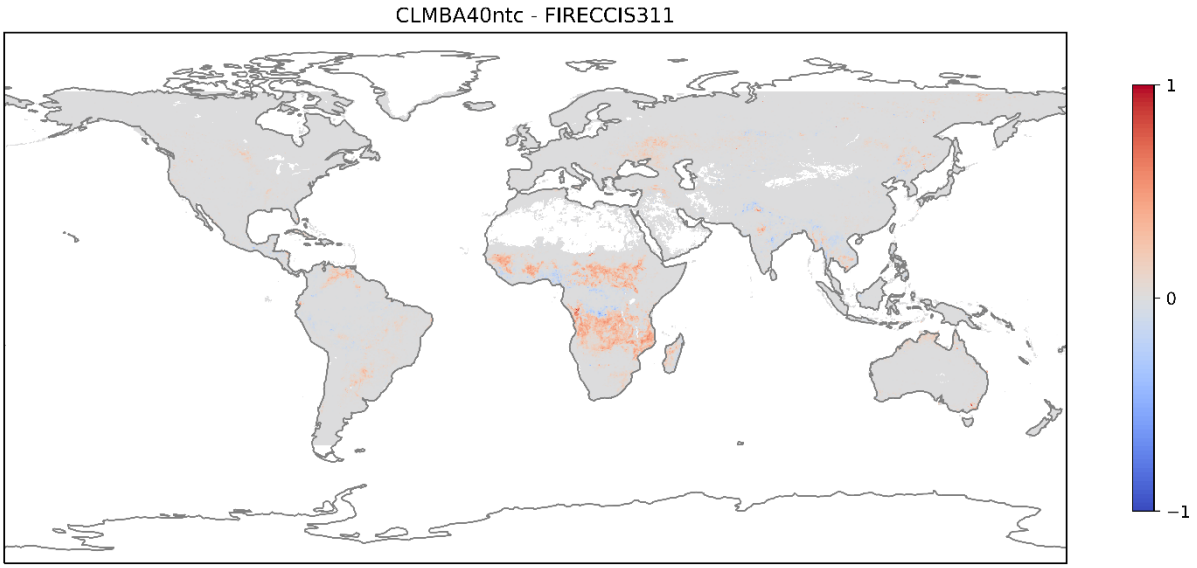


Figure 6: Difference map of proportion of burned area for 2020 between CLMBA40ntc and FIRECCIS311 at 0.25º spatial resolution.

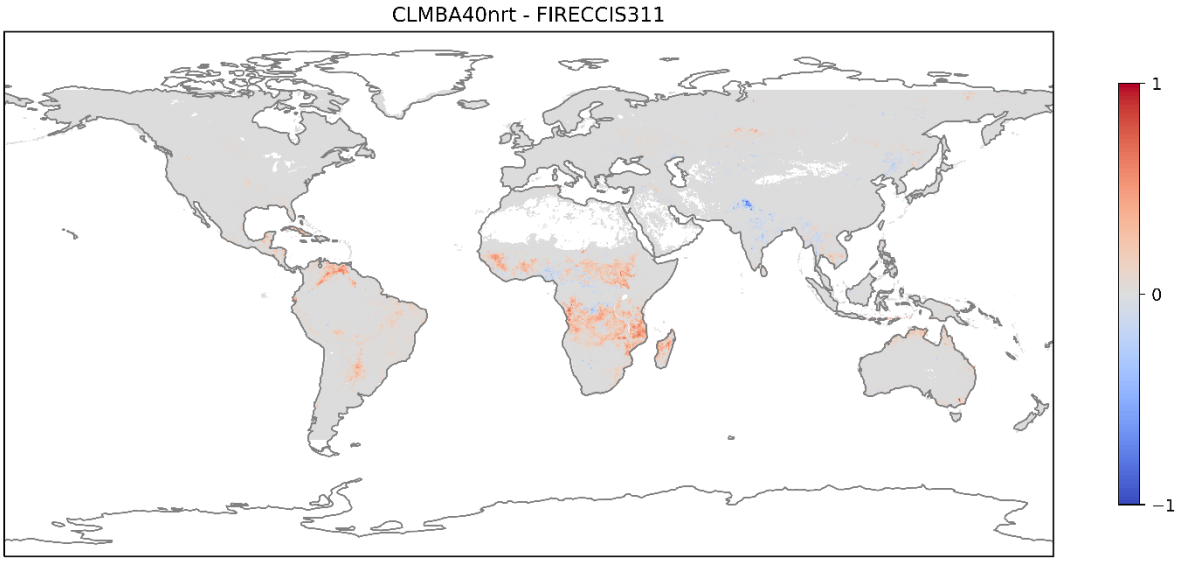


Figure 7: Difference map of proportion of burned area for 2020 between CLMBA40nrt and FIRECCIS311 at 0.25º spatial resolution.

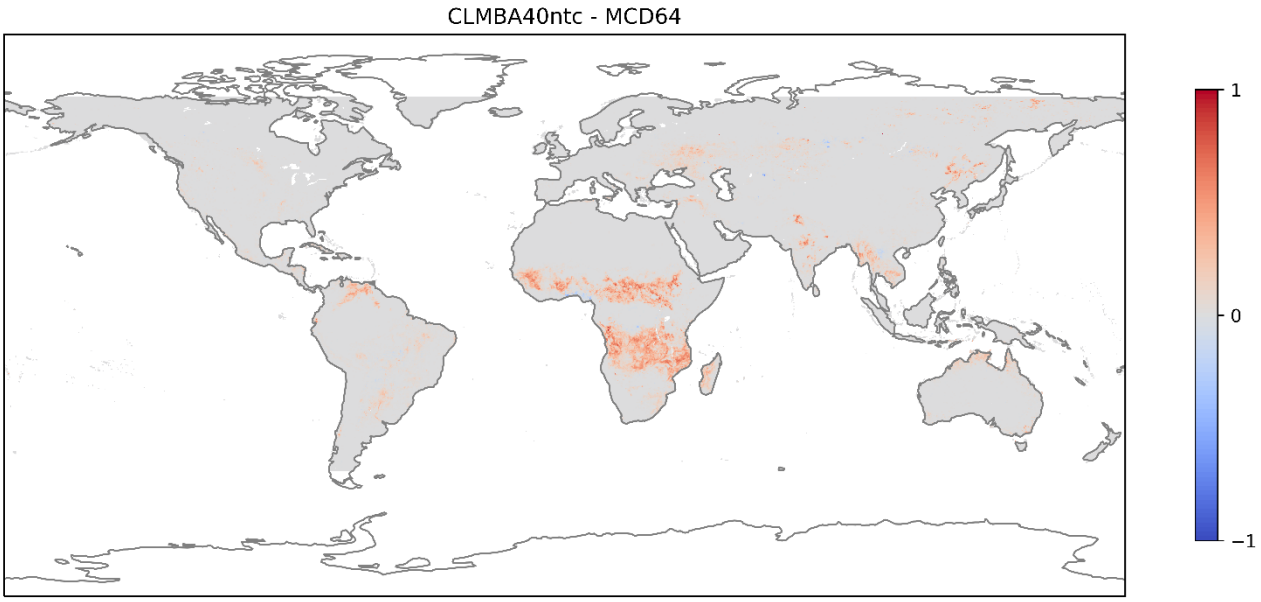


Figure 8: Difference map of proportion of burned area for 2020 between CLMBA40ntc and MCD64 at 0.25º spatial resolution.

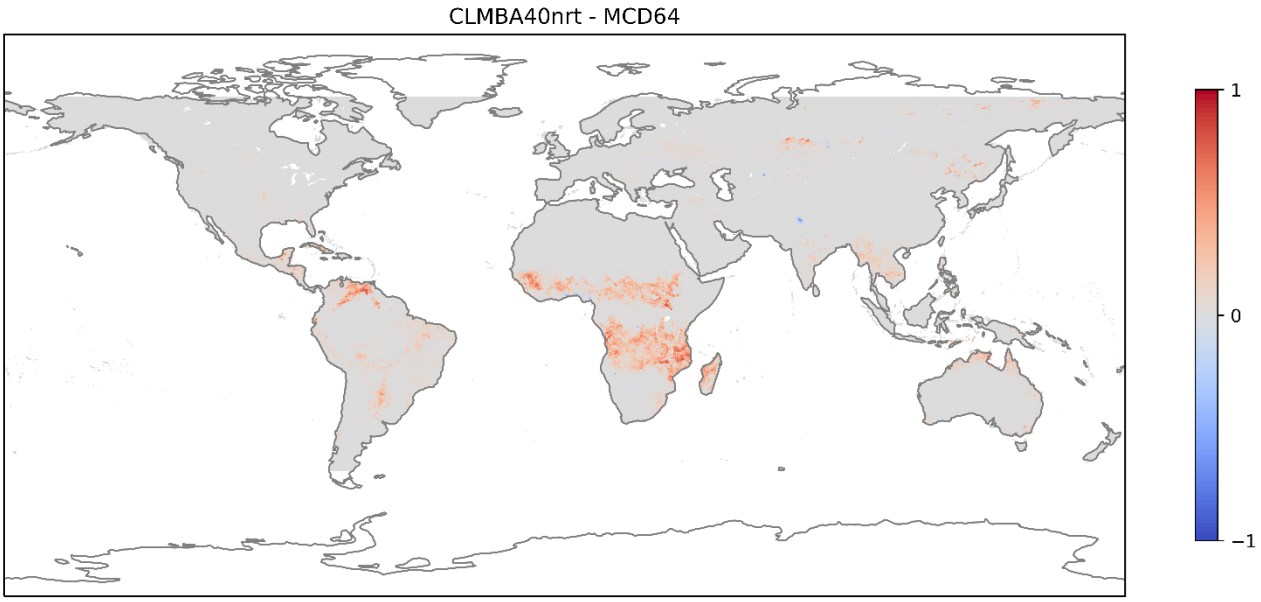


Figure 9: Difference map of proportion of burned area for 2020 between CLMBA40nrt and MCD64 at 0.25º spatial resolution.

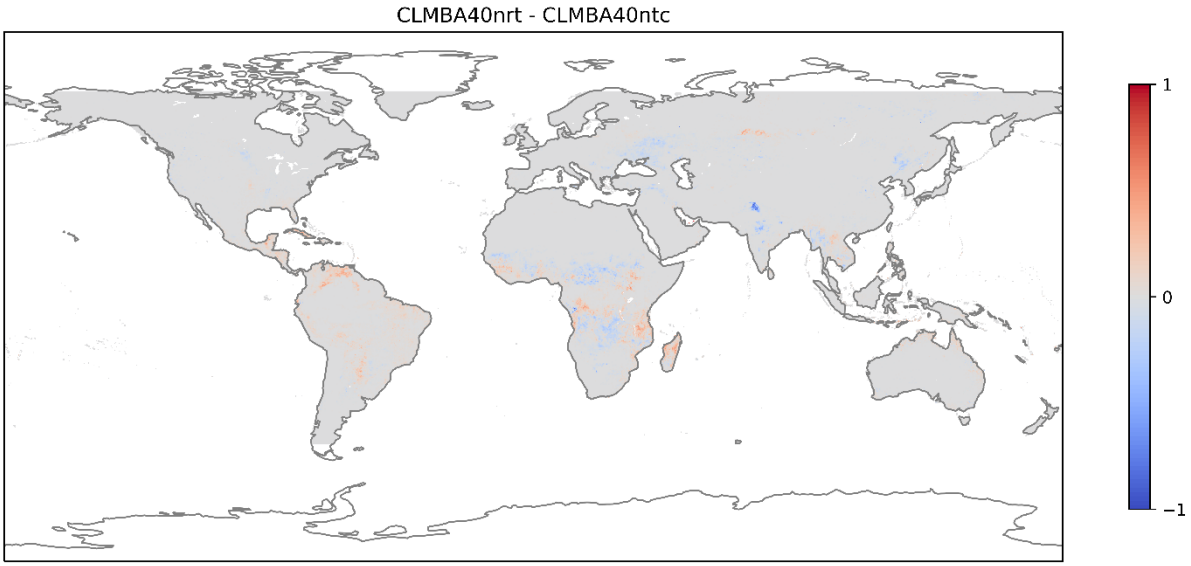


Figure 10: Difference map of proportion of burned area for 2020 between CLMBA40nrt and CLMBA40ntc at 0.25º spatial resolution.

Globally, the CLMBA40nrt detects slightly more burned area than CLMBA40ntc, only around the 4% more in 2020 (the year where both products are available; 694.1 Mha and 666.3 Mha, respectively). Furthermore, CLMS NTC product tends to detect around 40% (36% – 41%) more annual burned area than FIRECCIS311 and around 90% (83% - 94%) more than MCD64, quite systematically throughout 2019-2022 and 2019-2024, respectively (Table 3).

Table 3: Yearly totals of burned area (Mha) for years 2019 – 2024. "-" refers to incomplete or null coverage.

	CLMBA40nrt	CLMBA40ntc	FIRECCIS311	MCD64
2019	-	665.6	489.4	343.2
2020	695.3	666.5	473.2	348.1
2021	-	657.8	476.6	350.8
2022	-	622.7	441.4	326.3
2023	-	695.6	-	381.5
2024	-	706.0	-	364.2

Figure 11 shows the available monthly BA totals throughout January 2019 to August 2025 on each biome used in the stratification of the sampling design of the validation data. CLMBA40 and other products display the same seasonality over biomes where there is a seasonality in the fire occurrence

(for instance Tropical Savanna) but CLMBA40 product provide higher BA values than other products. CLMBA40ntc and CLMBA40nrt estimate remarkably similar amounts of BA through time particularly in Tropical Savana, the biome where most of the BA occur. CLMBA40nrt tends to estimate slightly more BA than CLMBA40ntc, mainly in Tropical regions with high fire activity, and consistently almost throughout all 2020 (the period where the two products' data overlaps). Conversely, CLMBA40ntc estimates more BA than CLMBA40nrt in Temperate Forest in November and in Deserts & Xeric Shrublands in May, which include the agricultural fires in North East India mentioned above. CLMBA40 products tend to estimate more BA than the other two non-CLMS products consistently through time and across biomes, particularly in the months where all products estimate high amount of BA. Another clear pattern is the high peaks of FIRECCIS311 in Tropical Forest in 2019, 2021 and 2022, which reflect the discrepancies mentioned above, possibly caused by small-scale deforestation.

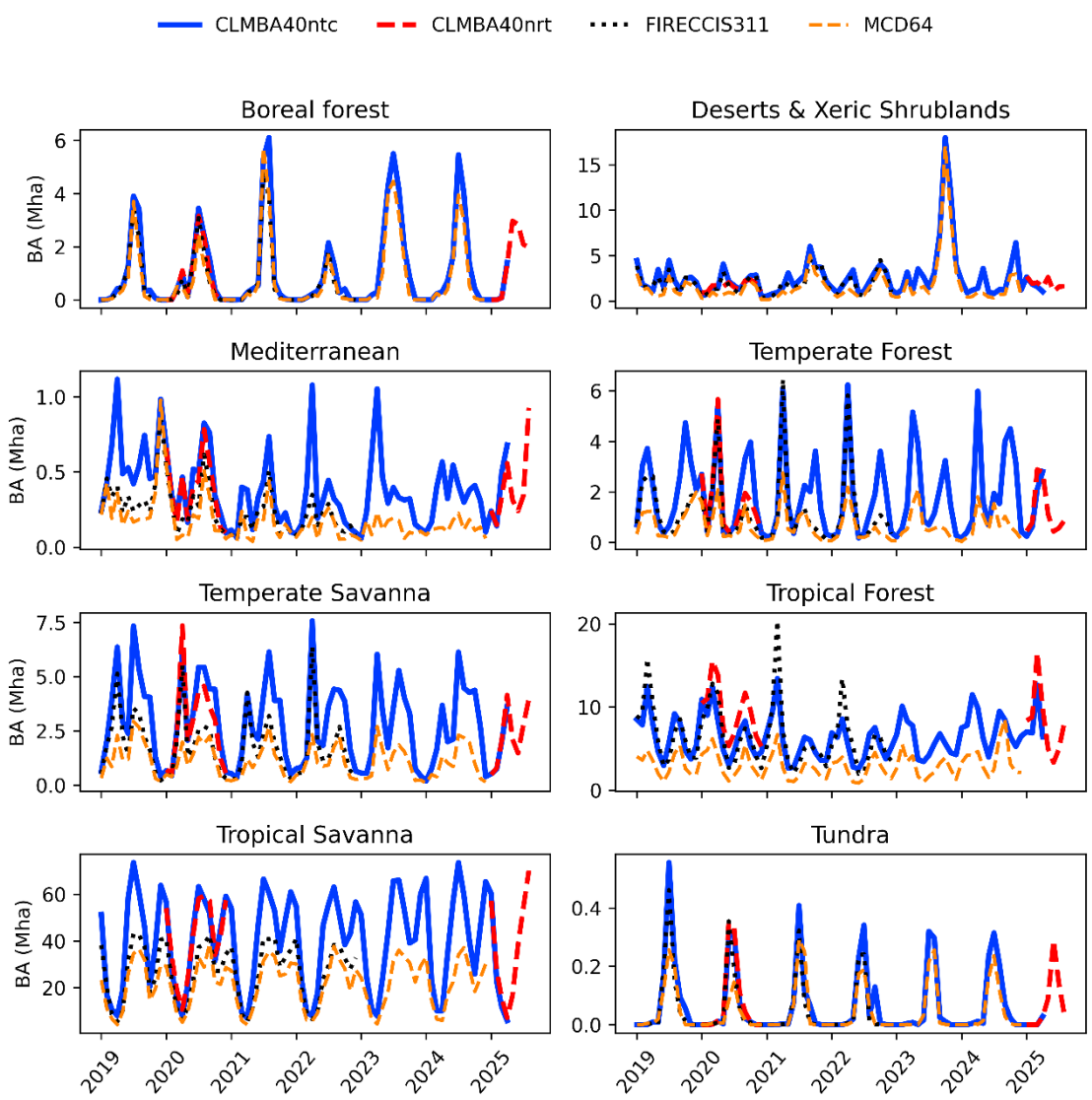


Figure 11: Temporal evolution of monthly burned area detected by each product throughout January 2019 – August 2025 and across biomes. For conciseness, biome Lakes and ice is not included.

The highest agreement with FIRECCIS311 is found for CLMBA40ntc according to the r^2, globally (r^2 0.846 in 2020), and with MCD64 according to the RMSE (RMSE 9.47 km^2) (Figure 12). Similar results are observed at biome level (Figures 13 and 14). CLMS products tend to estimate more BA than FIRECCIS311, particularly, in relative terms, for the lowest amounts of BA, as can be appreciated by the shape of the spreads of points around the ideal 1:1 line (dashed line in Figure 12). The lower RMSE values at the MCD64 vs. FIRECCIS311 comparison reflect the consistent higher BA estimates in the CLMS products.

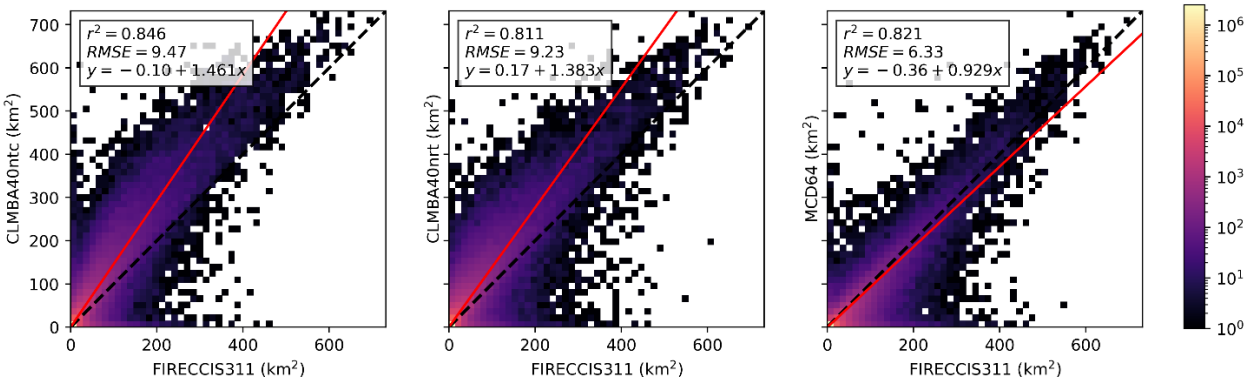


Figure 12: Scatter plots between the BA at the monthly 0.25º grid cells of assessed products, CLMBA40ntc, CLMBA40nrt and MCD64, and the BA of the FIRECCIS311 product for year 2020. The point density is shown with a logarithmic colour scale. The RMSE and regression results are displayed in the top-left text boxes and in red lines. The black dashed line represents the ideal 1:1 regression line.

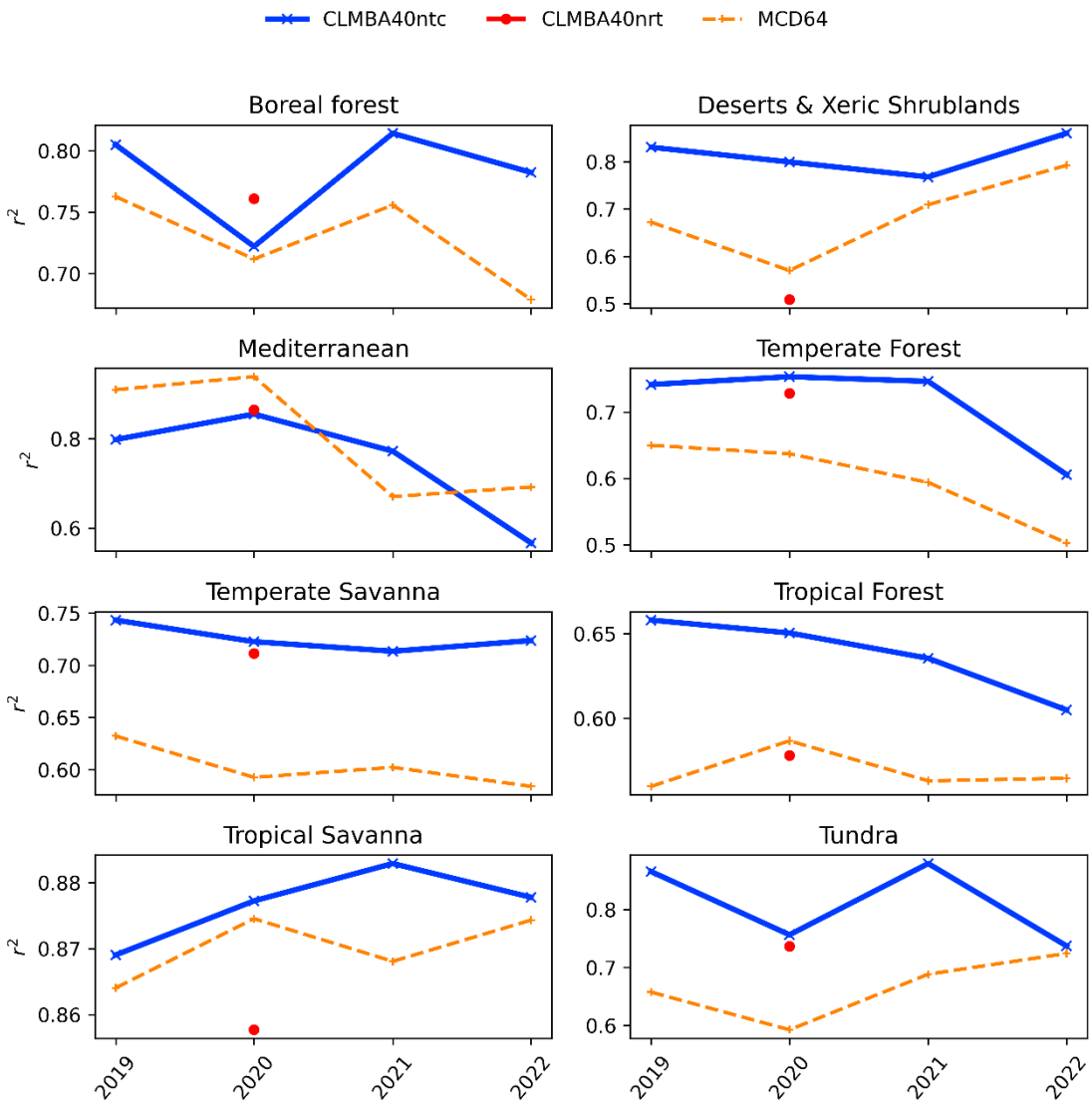


Figure 13: Per-biome temporal evolutions of the linear regression r^2 between the monthly 0.25° grid cells BA of CLMBA40ntc, CLMBA40nrt and MCD64 against FIRECCIS311. For conciseness, biome Lakes and ice is not included.

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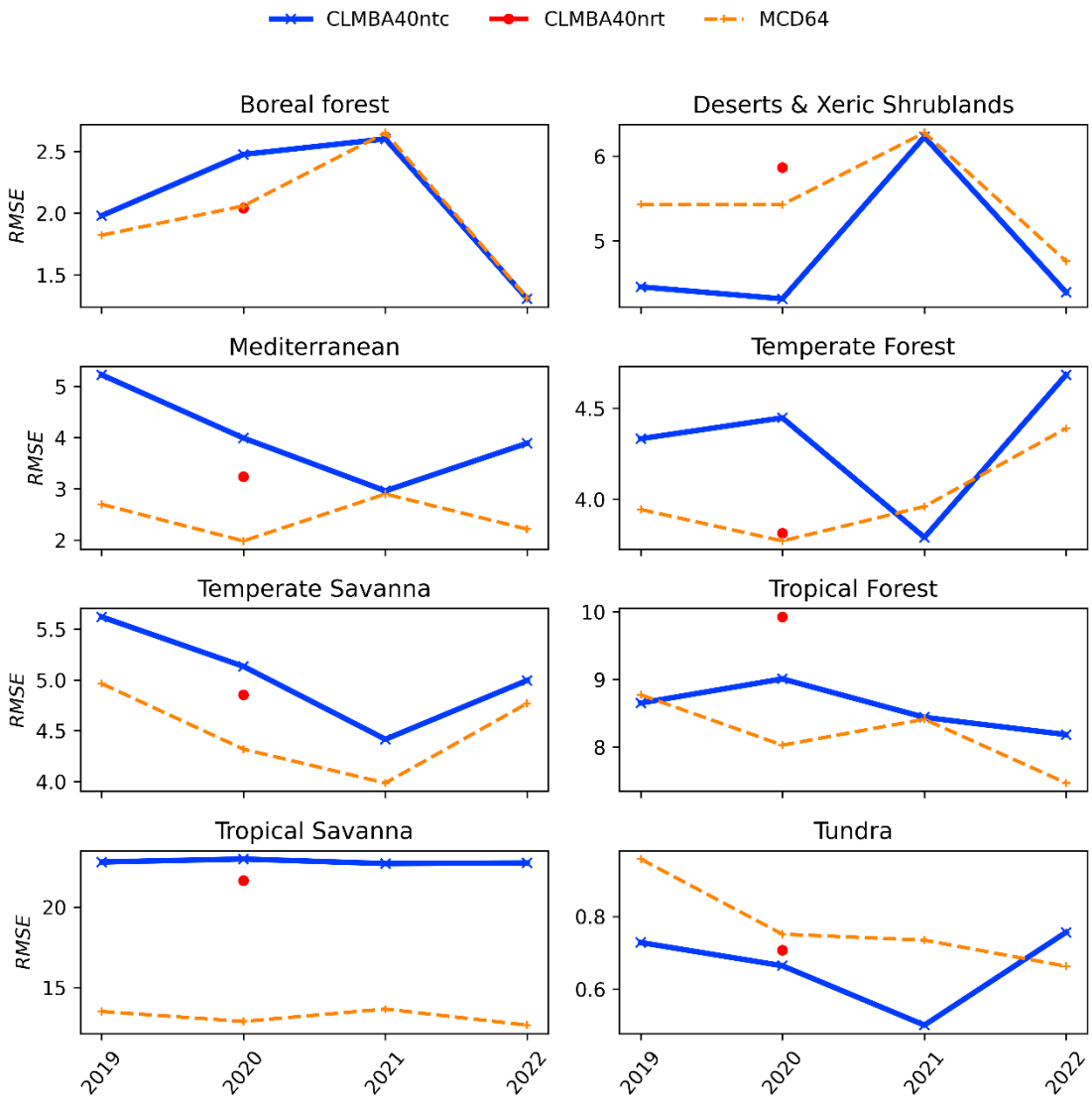


Figure 14: Per-biome temporal evolutions of RMSE of CLMBA40ntc, CLMBA40nrt and MCD64 BA estimates at the monthly 0.25º grid cells, taking as reference FIRECCIS311. For conciseness, biome Lakes and ice is not included.

4.2 VALIDATION ANALYSIS

Figure 15 and Table 4 present the accuracy metrics resulting of the validation of CLMS BA products against the reference dataset for the year 2020. For benchmarking, the MCD64 and FIRECCIS311 products are also validated similarly.

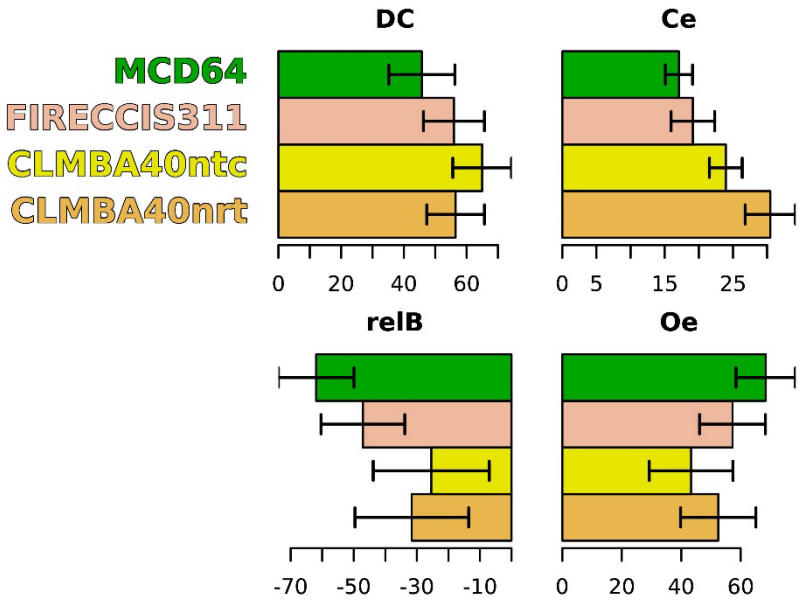


Figure 15: Estimated accuracy Dice coefficient (DC), Commission error ratio (Ce), Omission error ratio (Oe) and relative bias (relB), expressed as %, at the globe for the year 2020. Error bars express 95% confidence intervals

Globally, CLMBA40ntc is the most accurate product (Dice coefficient (DC) of 65.0%), followed by CLMBA40nrt (DC 56.5%) and FIRECCIS311 (DC 56.0%), and at a longer distance by MCD64 (DC 45.7%). Although the CLMS products have the highest commission error rates, they also have the lowest omission rates, leading to the lowest underestimation rates (relB ~-30%).

Table 4: Estimated accuracy Dice coefficient (DC), Commission error ratio (Ce), Omission error ratio (Oe) and relative bias (relB), expressed as %, at the globe for the year 2020. Standard errors are shown in parentheses.

Product	DC (%)	relB (%)	Ce (%)	Oe (%)
MCD64	45.7 (5.4)	-61.9 (6.1)	17.1 (1.0)	68.4 (5.1)
FIRECCIS311	56.0 (5.0)	-47.1 (6.8)	19.1 (1.6)	57.2 (5.6)
CLMBA40ntc	65.0 (4.8)	-25.4 (9.4)	24.0 (1.2)	43.3 (7.2)
CLMBA40nrt	56.5 (4.7)	-31.6 (9.2)	30.5 (1.9)	52.4 (6.5)

Figures 16 - 17 show accuracy measures DC and relB in each sampling unit (TSA) for CLMBA40ntc and CLMBA40nrt. Similar results for the other two products evaluated here are included in Annex 4. All products tend to show their respective highest accuracies at the same sampling units with high BA, e.g., in African Tropical Savana. This is expected as algorithms tend to map BA accurately in regions with large and compact BA (Padilla et al., 2015).

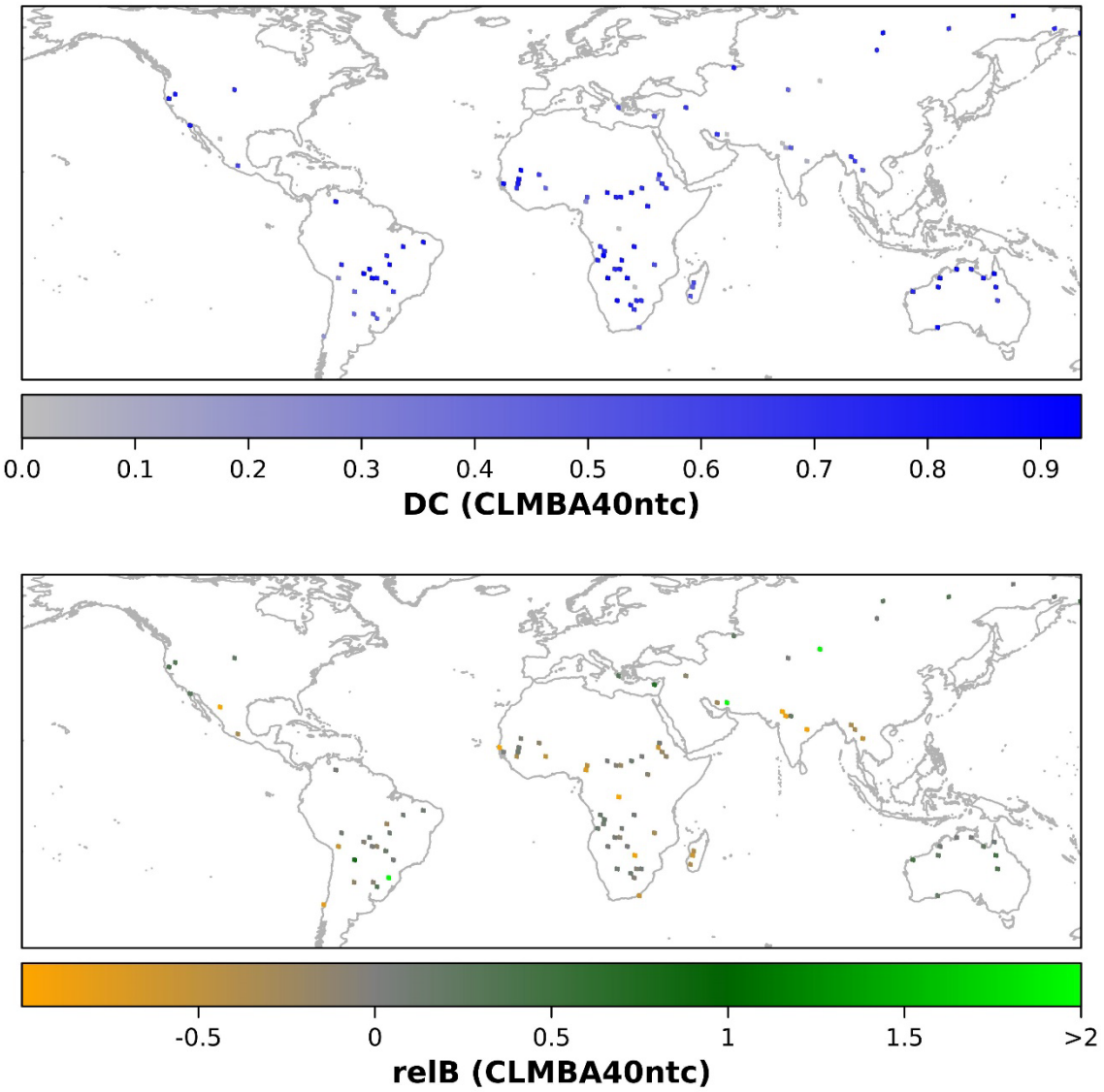


Figure 16: *DC* and *relB* on Thiessen Scene Area (TSAs) for CLMBA40ntc in 2020. TSAs with reference data but without accuracy measure available are not represented. For example, *DC* is not available when there is no BA in the reference data or in the product.

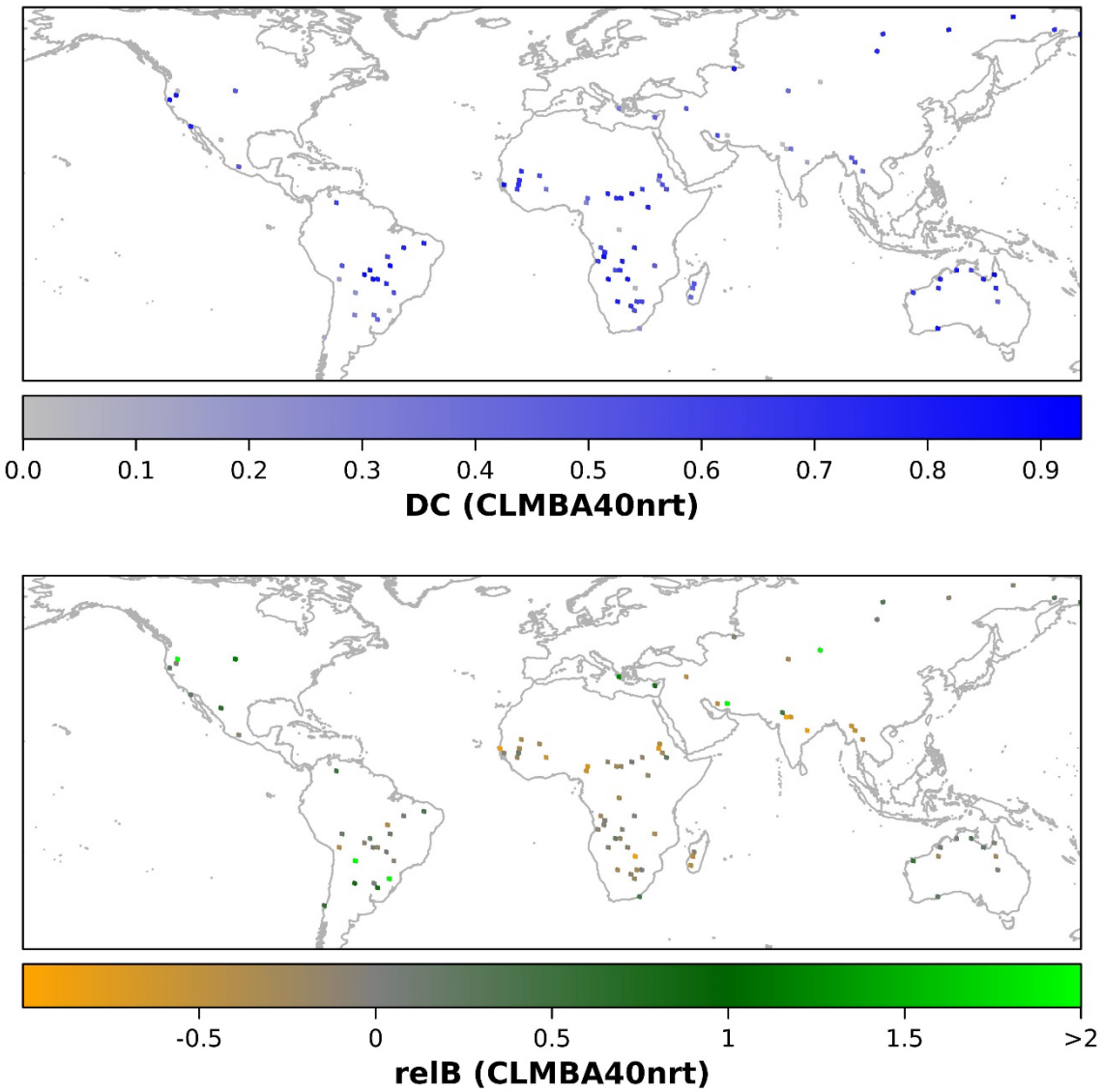


Figure 17: *DC* and *relB* on Thiessen Scene Area (TSAs) for CLMBA40nrt in 2020. TSAs with reference data but without accuracy measure available are not represented. For example, *DC* is not available when there is no BA in the reference data or in the product.

Annex 5 shows the estimated error matrices, at global and biome level.

5 DISCUSSIONS

It is remarkable how CLMBA40ntc is much more accurate than the other non-CLMS products (DC 65.0% vs up to 56.0%), and how CLMBA40nrt (DC 56.5%) has similar or even higher accuracies than the other non-CLMS products. It is important to remember that while **CLMBA40nrt is the only product derived at NRT** i.e. available for users within 24 hours after satellite acquisition, all other products are derived at a NTC mode.

CLMBA40ntc, as expected, has higher accuracy than CLMBA40nrt (DC 65.0% vs 56.5%), which is more affected by the noise of the input data, e.g. by residual cloud shadows not flagged by the Sentinel-3 quality layers. The use of CLMBA40nrt implies a sacrifice in terms of accuracy, 13% on DC compared with CLMBA40ntc, but at the benefit of a large reduction in the product's latency, from several months to hours.

The high accuracies of the CLMS products reflect the potential of the deep learning approach used here for BA mapping. This approach can be used with data from satellite missions other than Sentinel-3, for example, data at higher spatial resolution from Sentinel-2, as the training data is generated by sensor-independent semi-empirical models.

6 CONCLUSIONS AND PERSPECTIVES

Intercomparison analysis with other global satellite BA products throughout 2019-2024 and validation results, with Landsat-derived reference data in 2020, provide evidence of the products' quality and consistency on a multi-year basis. The accuracy of CLMBA40ntc (DC 65%) is higher than any other product. The accuracy of CLMBA40nrt (DC 56.5%) is similar than FIRECCIS311 (DC 56.0%), the most accurate non-CLMS product. Therefore, as expected, the accuracy of CLMBA40ntc is more accurate than CLMBA40nrt. The clear advantage of the later product, delivered at NRT, is that it provides BA estimates every day, within one day after the sensor acquisition, instead of several months after image sensing as in NTC products.

6.1 COMPLIANCE WITH REQUIREMENTS

All inter-compared products have omission error rates higher than 40%. Consequently, none meets the CLMS and GCOS threshold requirement of 25% (Chapter 2, GCOS#245). At the opposite, MCD64, FIRECCIS311 and CLMBA40ntc have commission errors lower than the 25% threshold requirement.

6.2 PERSPECTIVES

The sensor-independent capabilities of the current CLMS BA algorithm enable providing estimates at several timeliness and from several sensors. This can greatly facilitate the CLMS exploitation of other satellite missions, such as Sentinel-2, and better address end-user requirements.

7 RECOMMENDATIONS

The NRT product is recommended for applications that need burned area information shortly after the fire occurs. For non-time critical applications, the NTC product is recommended given its higher accuracy.

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ANNEX 1: MAPS OF AVERAGE BURNED AREA

This Annex shows the annually average of burned area across the period of availability of each NTC product: 2019 – 2024 for CLMBA40NTC and MCD64, 2019-222 for FIRECCIS311.

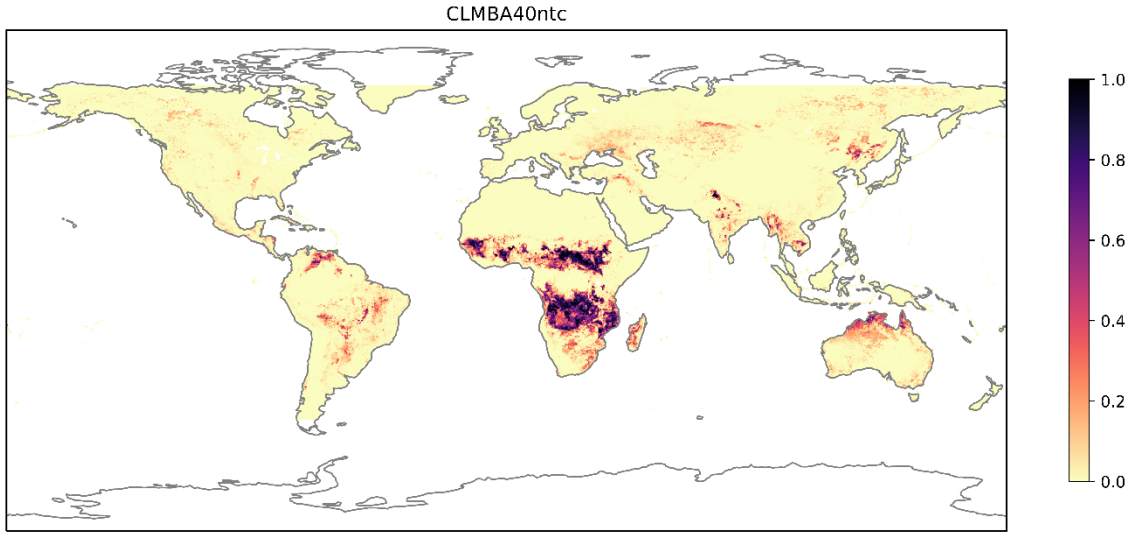


Figure 18: Mean of proportion of annual burned area in each 0.25º grid cell throughout 2019 – 2024, according to CLMBA40ntc. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell.

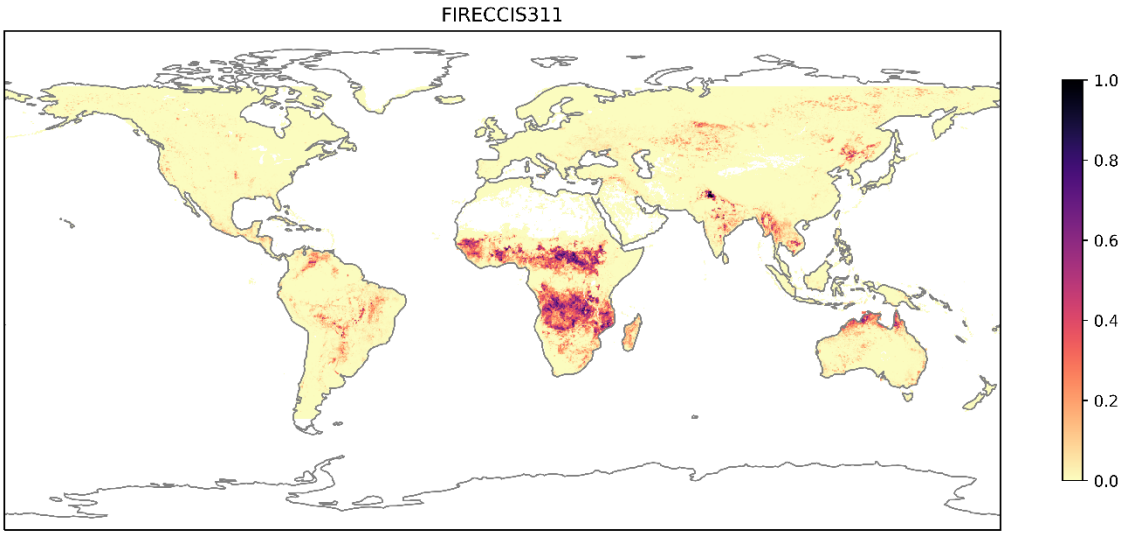


Figure 19: Mean of proportion of annual burned area in each 0.25º grid cell throughout 2019 - 2022 according to FIRECCIS311. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell. White additionally represents non-burnable areas (e.g., bare areas and permanent snow and ice) flagged similarly as permanent water in this product.

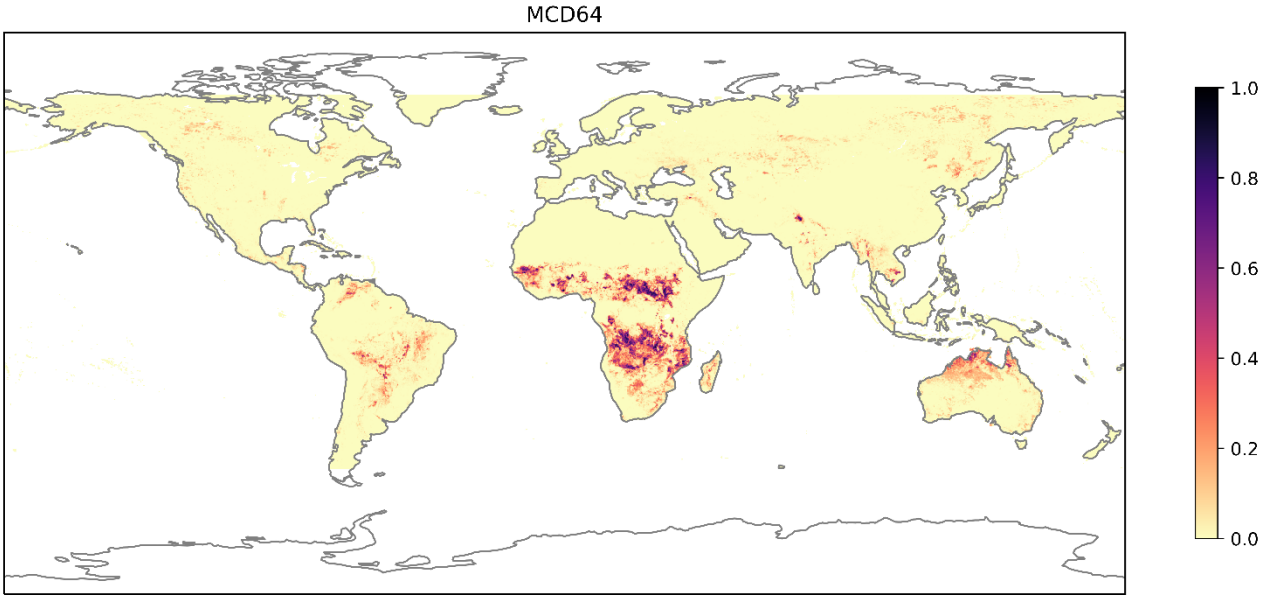


Figure 20: Mean of proportion of annual burned area in each 0.25º grid cell throughout 2019 - 2024 according to MCD64. White represents permanent water surfaces and not available estimations (no-data) in all pixels within the 0.25º grid cell.

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ANNEX 2: MONTHLY BURNED AREA MAPS FOR 2020

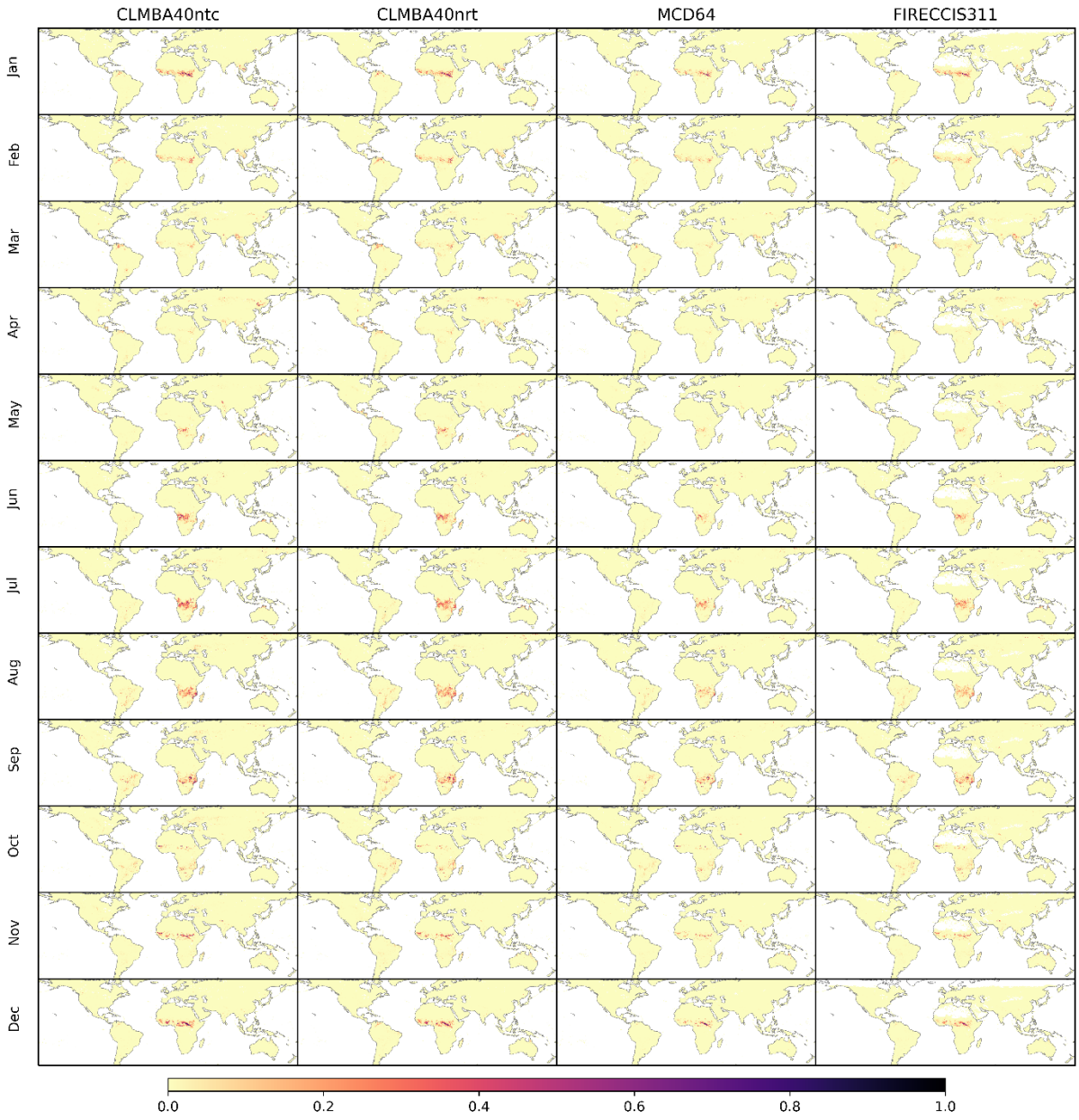


Figure 21: Proportion of burned area in each 0.25º grid cell for each month of 2020. White represents permanent water, not available estimations (no-data) in all pixels within the 0.25º grid cell and, additionally in FIRECCIS311, non-burnable areas (e.g. bare areas and permanent snow and ice).

ANNEX 3: COMPARISONS OF BURNED AREA DETECTIONS AND SENTINEL-3 OLCI & SLSTR IMAGES

On the following figures, Sentinel-3 images (SWIR, NIR and red bands at wavelengths 2.256, 0.865 and 0.665 µm, respectively, as RGB) for the two dates are shown in the top row. Black corresponds to unobserved pixels. Geographical coordinates are included as map labels on the x and y axes.

On Figure 22, relatively accurate estimates are provided by CLMBA40ntc over tropical savannah. It captures the burned patches slightly better than the other two products. CLMBA40nrt burned area is similar to CLMBA40ntc, although it also wrongly includes some residual cloud shadows (not detected by the Sentinel-3 quality layers).

Figure 23 illustrates a challenging region, over agricultural fields in South Asia, where errors occurred in all products. These errors primarily consisted of commission errors in CLMBA40ntc and FIRECCIS311, and omission errors in CLMBA40nrt. The mosaic of agricultural fields leads to land surface changes at the sub-pixel level, with a small impact on the Sentinel-3 300 m reflectance observations. Further exploration with Sentinel-2 imagery (not shown here) reveals the extreme difficulty in discriminating between harvesting or ploughing and burned area. While some agricultural fields can be unequivocally associated with burned area, due to the presence of fire fronts, many other fields present reflectance changes and spatial textures compatible with harvesting or ploughing.

Figure 24 shows false detections mainly in FIRECCIS311 and also in CLMBA40nrt over tropical forest in Africa, between the 14th of June and 4th of July 2020. While relatively large clusters of VIIRS active fires are detected (not shown in this figure), land surface changes are not observed at 300 m reflectance observations. Further exploration with Sentinel-2 imagery (not shown) revealed apparent fire-related reflectance changes at 20 m observations but not at the locations of the products' detections.

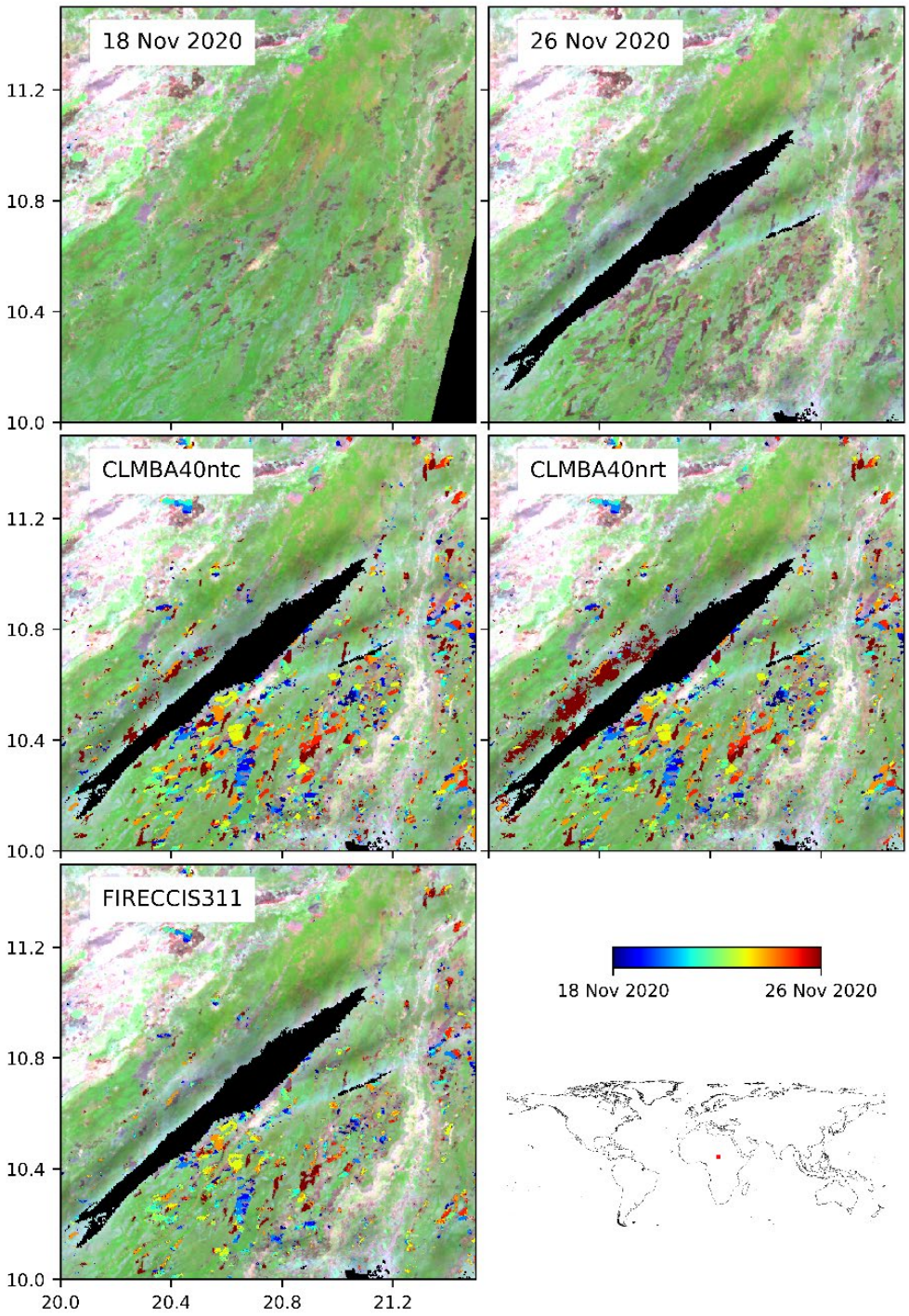


Figure 22: Burned area detected between the 18th and 26th of November 2020 by CLMBA40ntc, CLMBA40nrt and FIRECCIS311, over tropical savannah in central Africa. The 26th November 2020 image and the burned area detections are shown in the middle row and the bottom left panel.

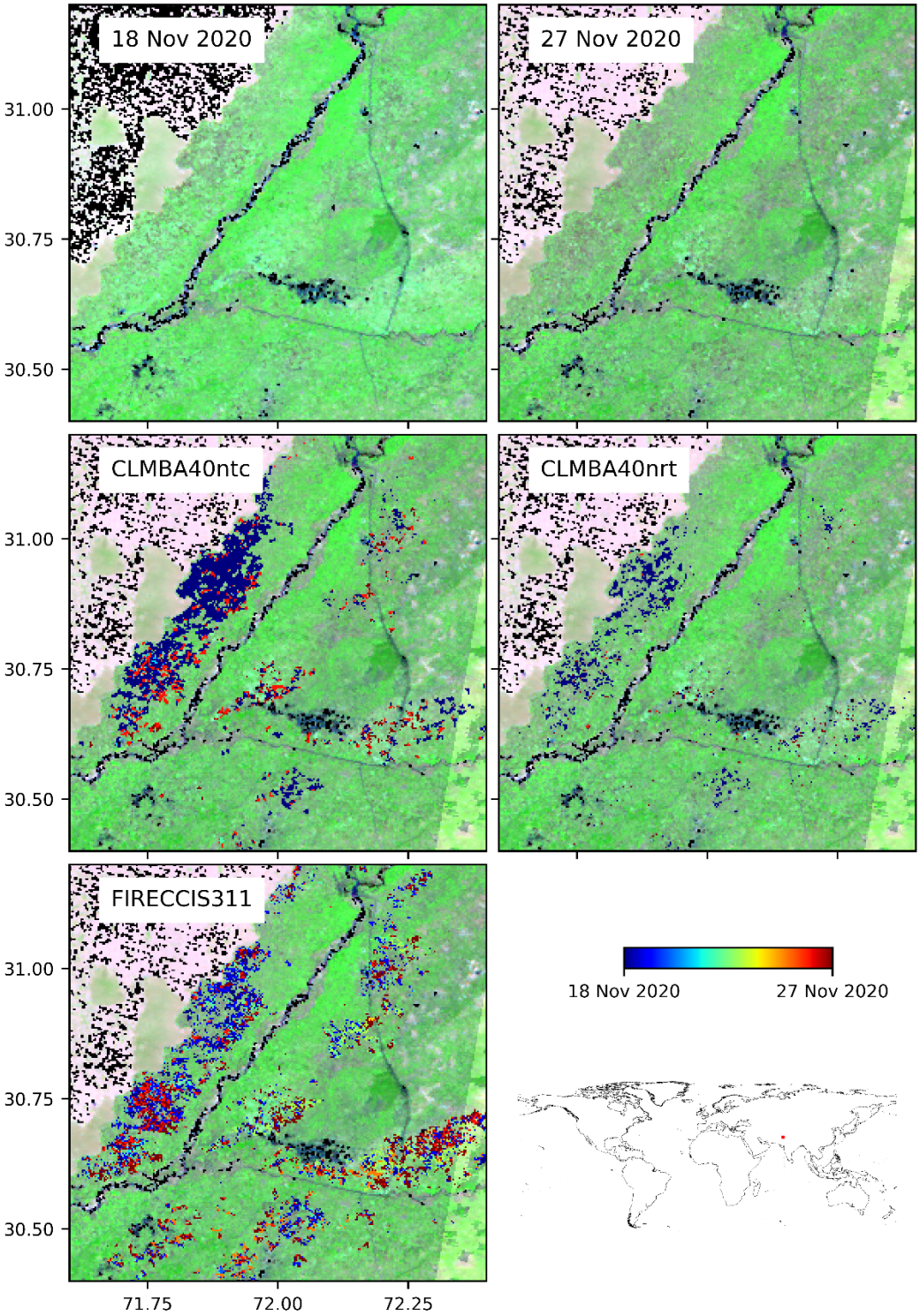


Figure 23. Burned area detected between November 18th and 27th, 2020 over agricultural fields in South Asia

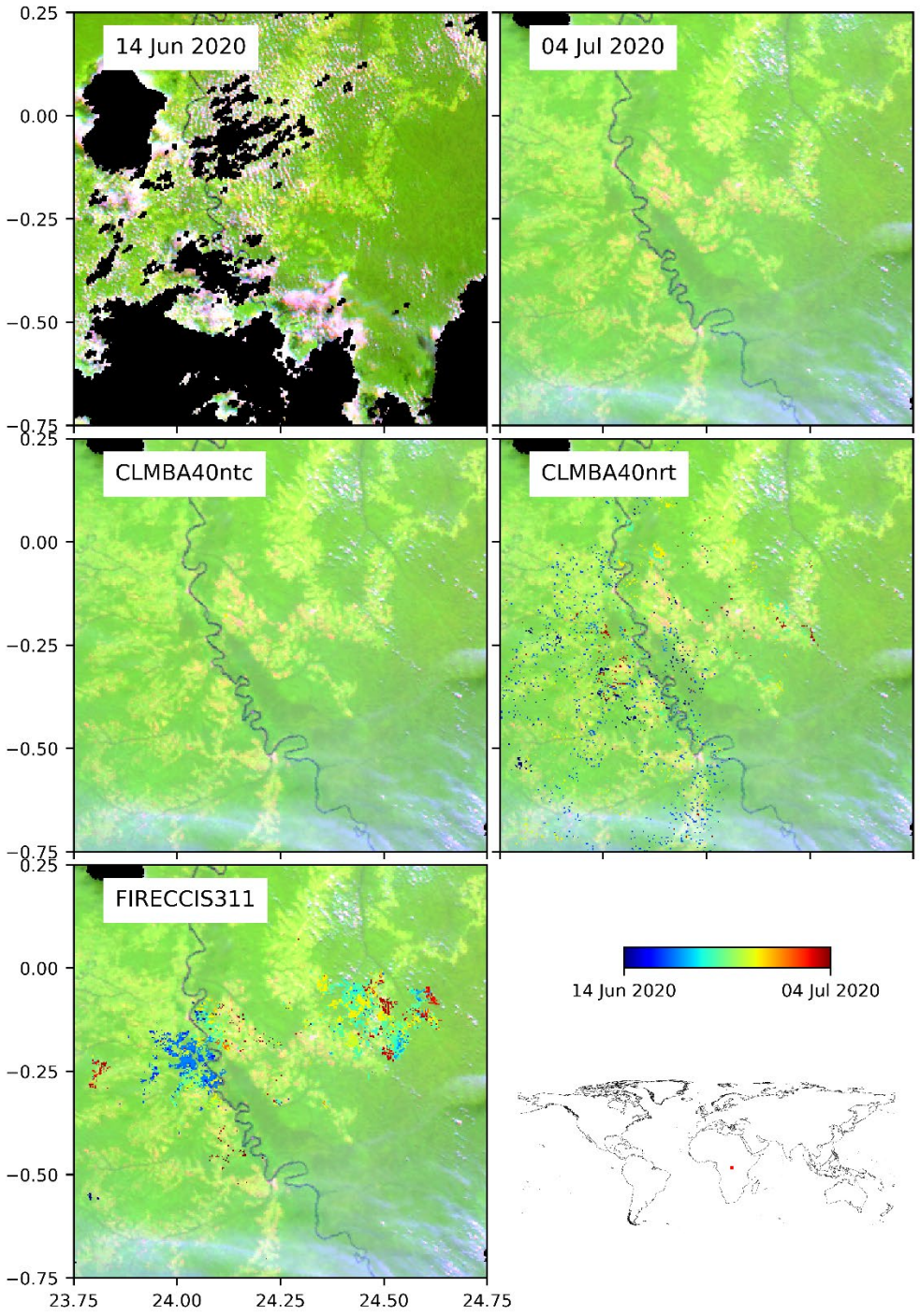


Figure 24. Burned Areas detected between the 14th of June and 4th of July 2020 over tropical forest in Africa.

ANNEX 4: ACCURACY RESULTS AT SAMPLING UNITS' LEVEL

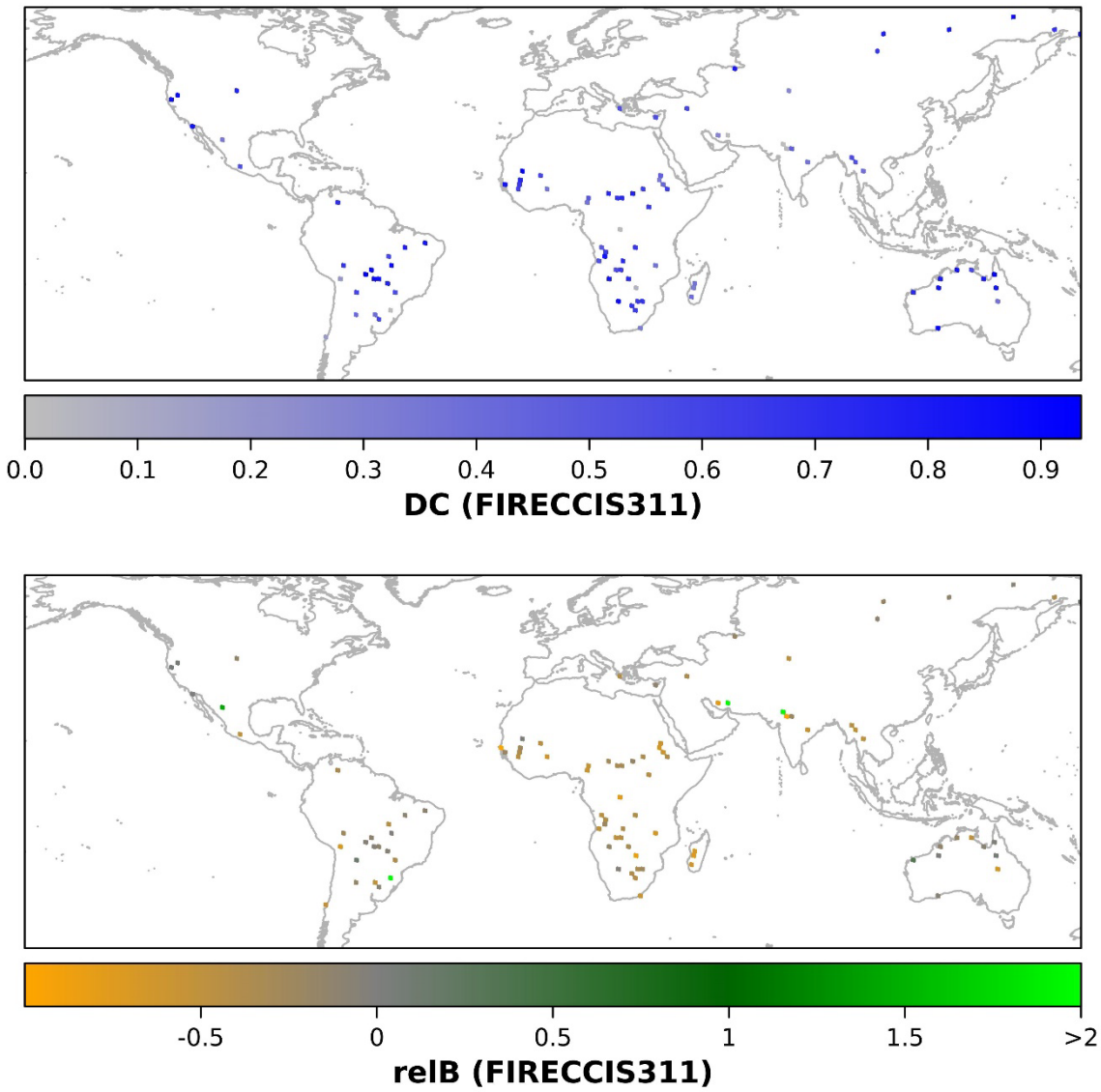


Figure 25: DC and relB on Thiessen Scene Area (TSAs) for FIRECCIS311 in 2020. TSAs with reference data but without accuracy measure available are not represented. For example, DC is not available when there is no BA in the reference data or in the product.

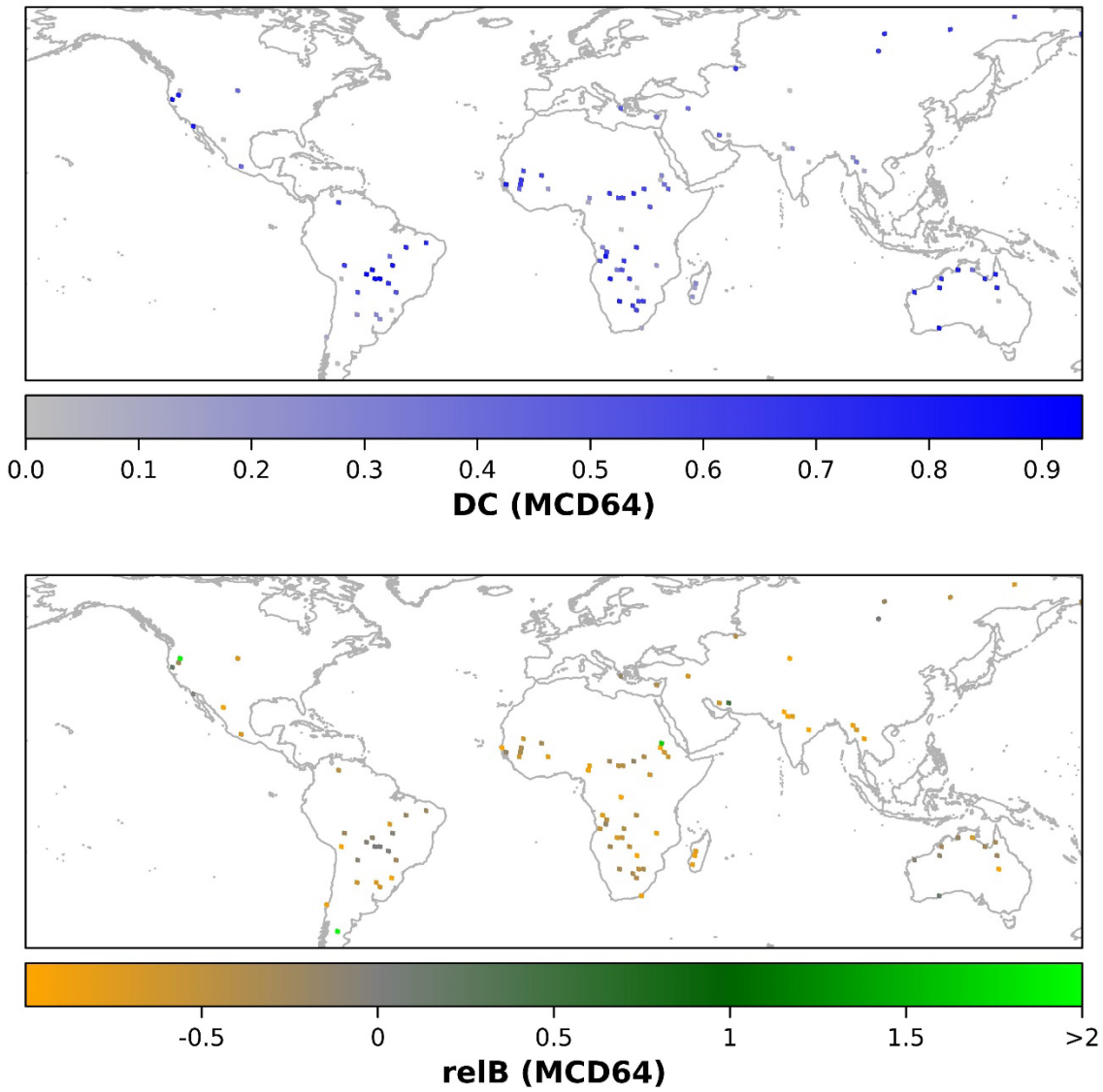


Figure 26: DC and relB on Thiessen Scene Area (TSAs) for MCD64 in 2020. TSAs with reference data but without accuracy measure available are not represented. For example, DC is not available when there is no BA in the reference data or in the product.

ANNEX 5: ESTIMATED ERROR MATRICES

Table 5: Global error matrices for 2020. Standard errors are shown in parentheses.

Product	E_{11} (km^2)	E_{12} (km^2)	E_{21} (km^2)	E_{22} (km^2)
MCD64	231.0 (32.3)	47.6 (7.2)	500.3 (113.0)	13510.9 (1990.8)
FIRECCIS311	312.7 (39.6)	73.9 (12.0)	418.1 (103.9)	13485.1 (1857.8)
CLMBA40ntc	412.1 (52.6)	129.8 (18.5)	314.6 (96.9)	13433.3 (1861.9)
CLMBA40nrt	347.5 (43.6)	152.2 (21.4)	383.1 (103.5)	13407.0 (1846.1)

Table 6: Per-biome error matrices for MCD64 in 2020. Standard errors are shown in parentheses.

Biome	E_{11} (Mha)	E_{12} (Mha)	E_{21} (Mha)	E_{22} (Mha)
Boreal forest	3.87 (2.18)	0.34 (0.07)	5.33 (3.16)	1147.33 (373.05)
Deserts & Xeric Shrublands	11.14 (7.65)	2.11 (1.11)	23.16 (17.75)	2781.78 (1275.17)
Mediterranean	3.96 (2.24)	1.09 (0.60)	0.89 (0.40)	275.94 (100.40)
Temperate Forest	20.50 (18.68)	5.63 (5.22)	7.93 (4.30)	1752.26 (934.38)
Temperate Savanna	10.14 (5.95)	2.86 (1.45)	15.10 (6.54)	1785.60 (481.51)
Tropical Forest	27.84 (9.78)	6.28 (2.03)	238.51 (103.37)	3213.51 (735.53)
Tropical Savanna	153.43 (22.27)	29.21 (4.03)	209.24 (41.16)	1889.90 (188.75)
Tundra	0.13 (0.13)	0.07 (0.07)	0.15 (0.15)	664.59 (711.57)

Table 7: Per-biome error matrices for FIRECCIS311 in 2020. Standard errors are shown in parentheses.

Biome	E_{11} (Mha)	E_{12} (Mha)	E_{21} (Mha)	E_{22} (Mha)
Boreal forest	6.3 (3.84)	0.9 (0.53)	1.8 (1.15)	1148.0 (316.36)
Deserts & Xeric Shrublands	17.3 (10.99)	4.4 (2.63)	17.1 (14.59)	2779.4 (1232.49)
Mediterranean	4.0 (2.20)	0.6 (0.27)	0.8 (0.34)	276.5 (100.15)
Temperate Forest	20.3 (17.45)	5.6 (3.74)	8.1 (4.14)	1752.4 (932.54)
Temperate Savanna	12.1 (6.17)	3.0 (1.38)	13.0 (5.98)	1785.5 (481.82)
Tropical Forest	61.8 (19.42)	24.7 (9.38)	204.9 (97.01)	3194.7 (733.24)
Tropical Savanna	190.5 (26.54)	34.6 (5.79)	172.3 (33.42)	1884.4 (188.77)
Tundra	0.4 (0.08)	0.1 (0.05)	0.2 (0.12)	664.2 (383.62)

Table 8: Per-biome error matrices for CLMBA40ntc in 2020. Standard errors are shown in parentheses.

Biome	E_{11} (Mha)	E_{12} (Mha)	E_{21} (Mha)	E_{22} (Mha)
Boreal forest	4.8 (3.54)	1.2 (0.58)	0.5 (0.22)	1150.3 (249.41)
Deserts & Xeric Shrublands	22.0 (13.42)	7.2 (4.32)	12.3 (11.04)	2776.7 (1247.70)
Mediterranean	4.5 (2.37)	1.5 (0.66)	0.3 (0.12)	275.6 (99.63)
Temperate Forest	23.6 (19.45)	9.5 (7.41)	4.7 (3.46)	1748.5 (941.10)
Temperate Savanna	17.4 (9.09)	8.3 (3.76)	7.7 (3.45)	1780.3 (480.83)
Tropical Forest	75.2 (23.78)	26.6 (7.91)	191.1 (93.42)	3193.3 (734.23)
Tropical Savanna	264.0 (39.25)	75.3 (13.81)	98.0 (22.66)	1844.5 (187.08)
Tundra	0.5 (0.10)	0.2 (0.05)	0.1 (0.07)	664.1 (382.95)

Table 9: Per-biome error matrices for CLMBA40nrt in 2020. Standard errors are shown in parentheses.

Biome	E_{11} (Mha)	E_{12} (Mha)	E_{21} (Mha)	E_{22} (Mha)
Boreal forest	7.2 (4.02)	1.2 (0.58)	2.0 (1.20)	1146.5 (248.73)
Deserts & Xeric Shrublands	15.8 (9.52)	6.5 (4.93)	18.5 (13.73)	2777.4 (1231.75)
Mediterranean	4.1 (2.25)	1.9 (0.77)	0.6 (0.29)	275.2 (99.45)
Temperate Forest	20.7 (17.83)	10.8 (7.73)	7.6 (4.00)	1747.2 (932.03)
Temperate Savanna	14.0 (7.16)	9.3 (4.26)	11.2 (4.94)	1779.2 (480.46)
Tropical Forest	60.8 (19.80)	35.4 (12.76)	205.4 (98.31)	3184.5 (733.70)
Tropical Savanna	224.4 (32.08)	86.8 (13.89)	137.7 (28.58)	1833.0 (185.94)
Tundra	0.5 (0.11)	0.3 (0.07)	0.1 (0.05)	664.1 (382.88)

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